

# Box Robot Base

version 1.0, rev 15 27 December 2025

This is a classic box robot design intended to provide two forward facing attachments or a combination of forward and sidefacing attachments. We usually configure the left attachment motor for our worm drive attachment and the right attachment motor, rotated at 90 degrees, is usually for geared lifting arm. We also use the right motor to drive a forward facing axles to turn objects on the FLL board (like the gears in Masterpiece Mission 12 rotating the chicken or Unearthed Mission 11/Angler Artifact).

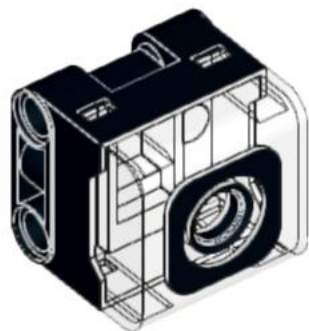
When assembling the robot care needs to be taken on how the sensor and motor cables need to be routed prior to joining the front and rear sections. Then secure the cables using the wire connector/49283 in rear compartment containing the hub.

It is loosely based on:

<https://www.youtube.com/watch?v=PCMEvf4p2PI&t=2s>



116x  
2780  
Black



2x  
37308c01  
Black



2x  
56908  
Black



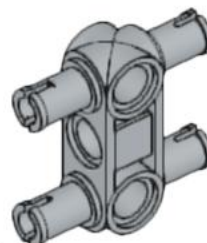
5x  
73507  
Black



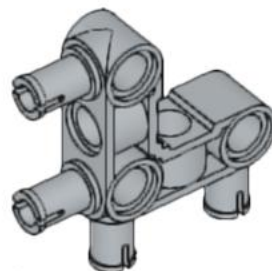
2x  
32073  
Black



4x  
32525  
Black

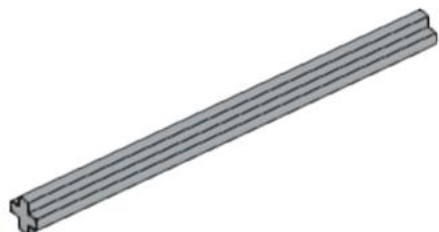


2x  
48989  
Light Bluish Gray

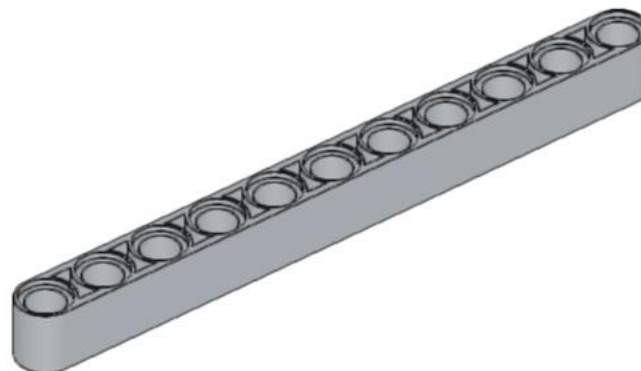


2x  
55615  
Light Bluish Gray

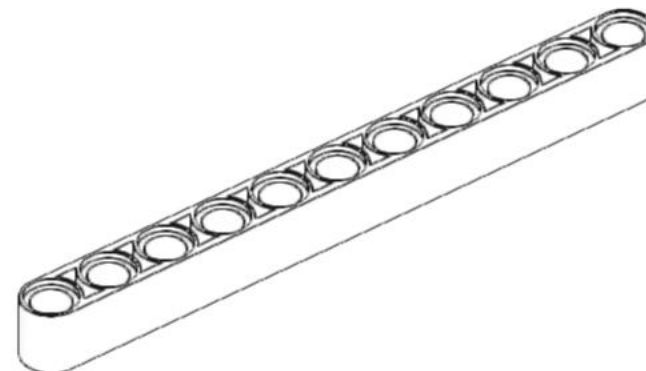
You might want to use a 4L axle instead of the 5L.



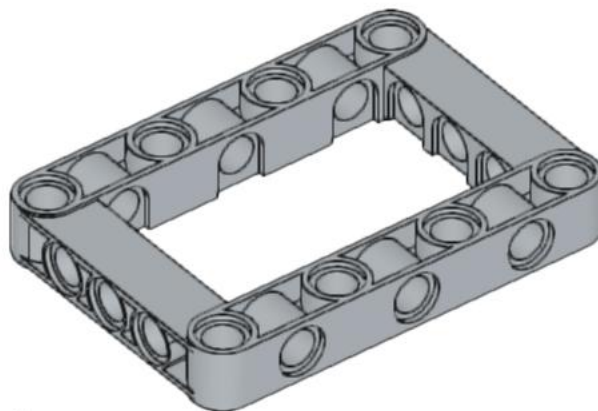
2x  
44294  
Light Bluish Gray



3x  
32525  
Light Bluish Gray



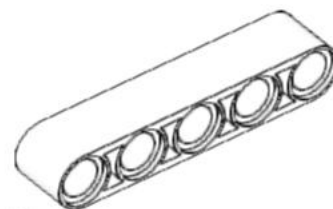
1x  
32525  
White



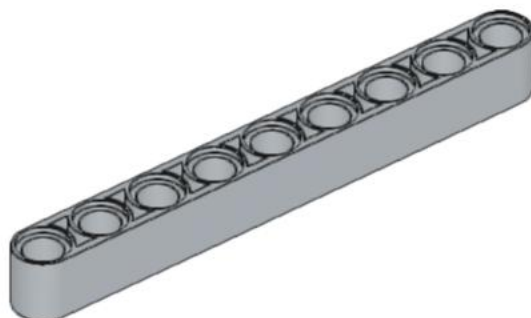
4x  
64179  
Light Bluish Gray



4x  
18654  
White



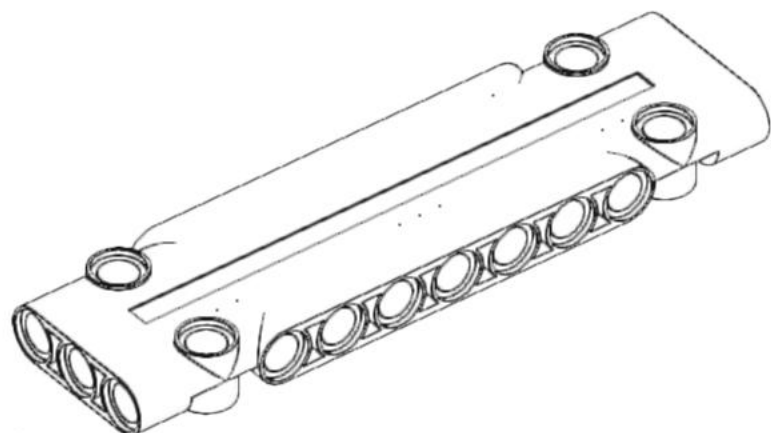
2x  
32316  
White



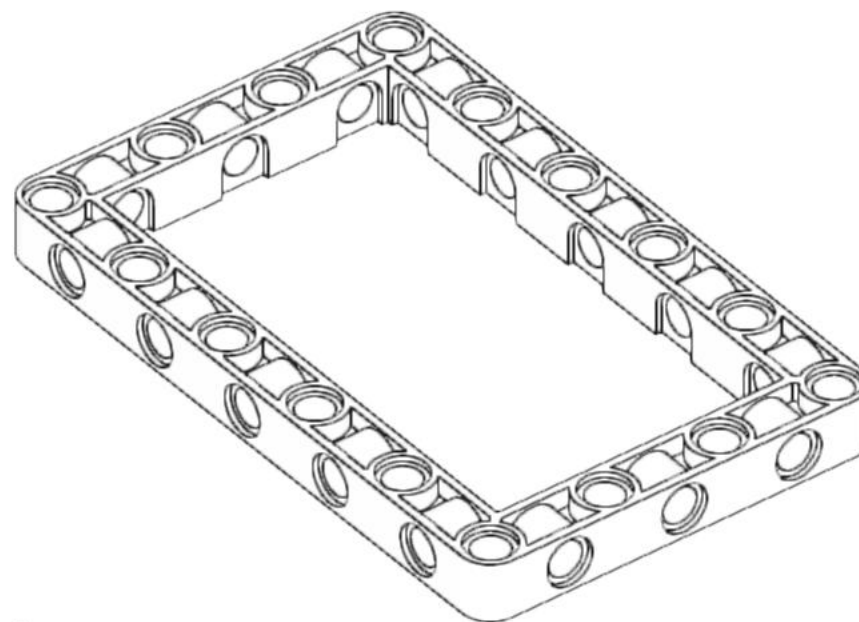
8x  
40490  
Light Bluish Gray



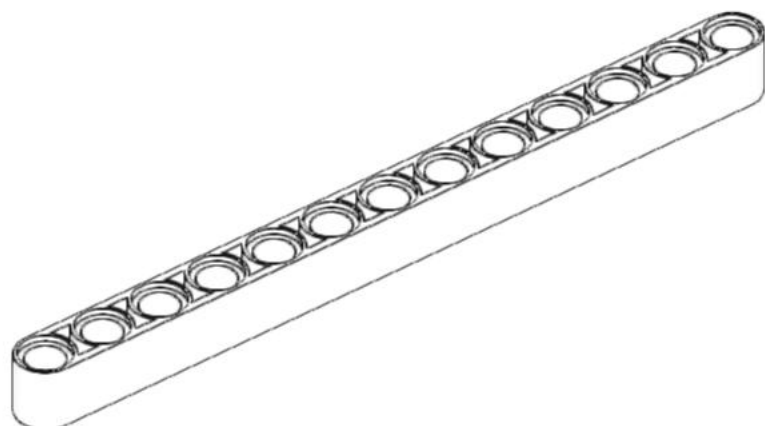
2x  
32020c01  
White



**2x**  
**15458**  
**White**

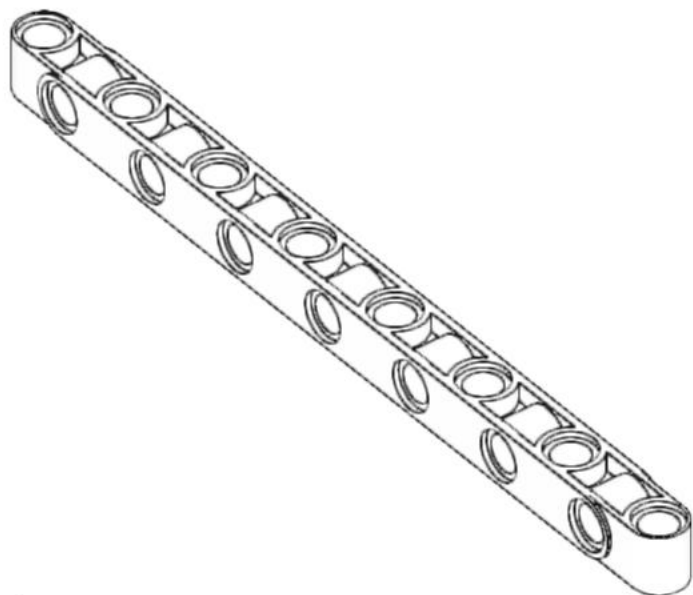


**3x**  
**39794**  
**White**

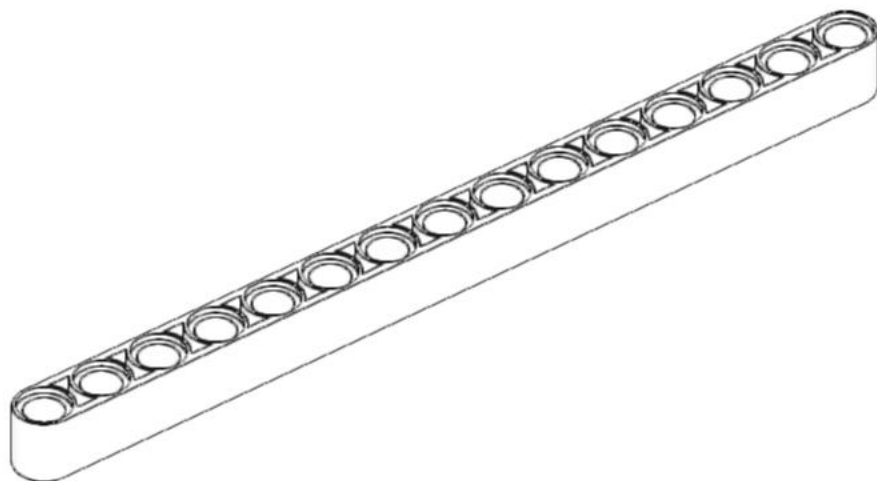


**2x**  
**41239**  
**White**

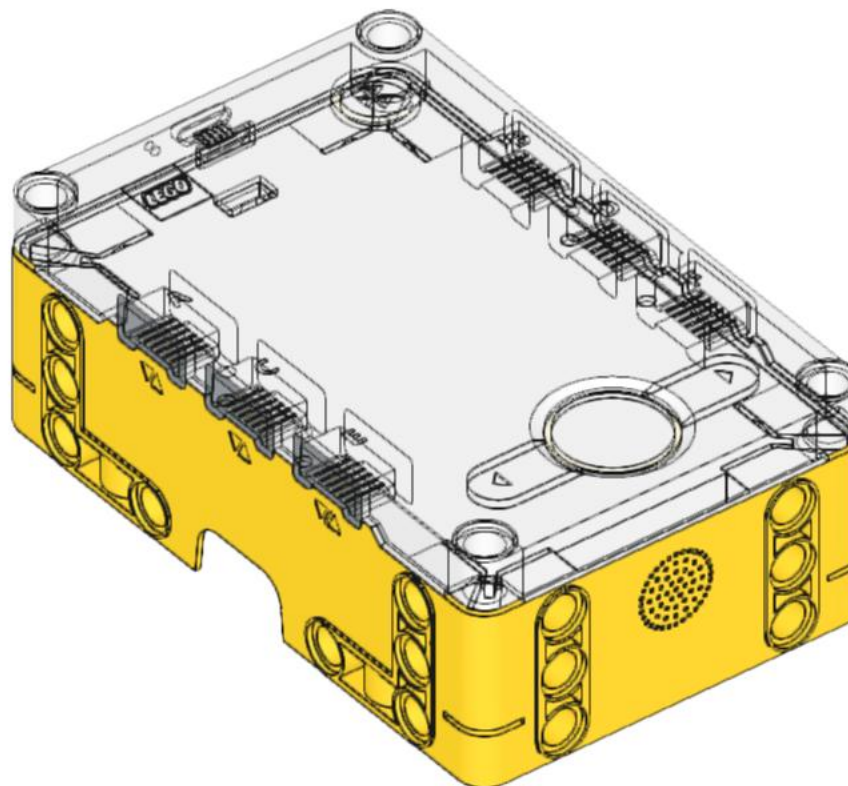




1x  
71710  
White



1x  
32278  
White



1x  
53444c01  
Yellow



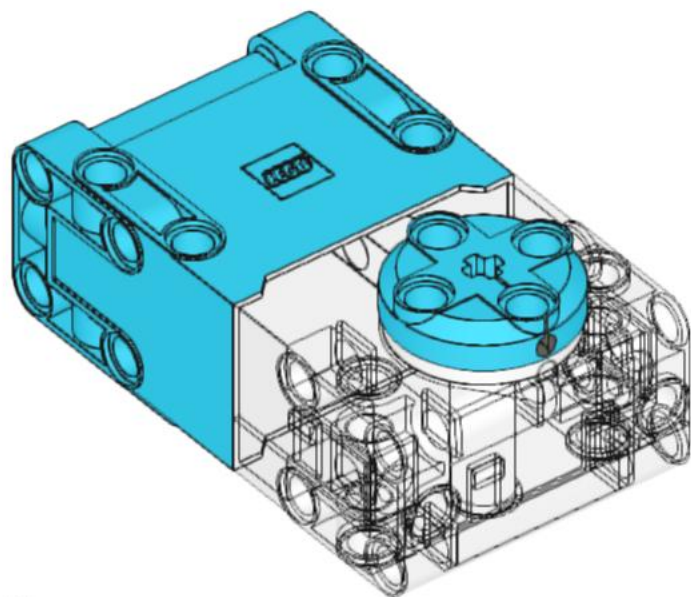
5x  
39369  
Yellow



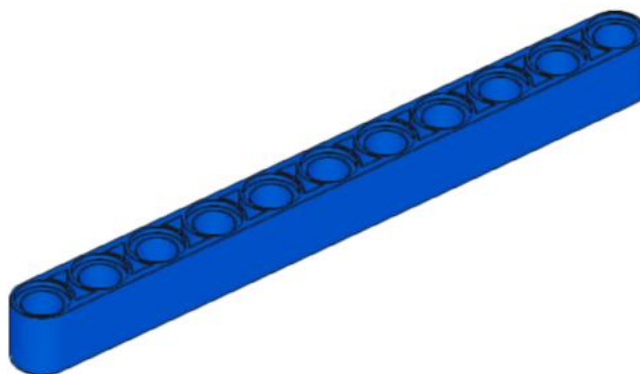
2x  
4265c  
Blue



81x  
6558  
Blue



2x  
54675c01  
Medium Azure



2x  
32525  
Blue

## Front Assembly

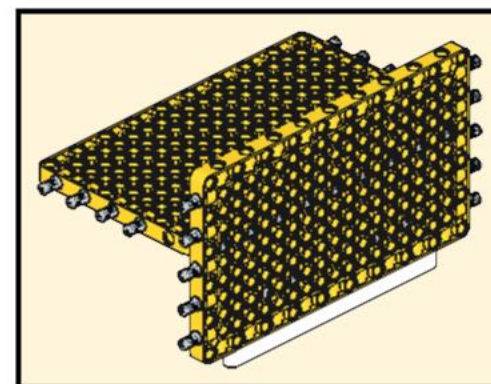
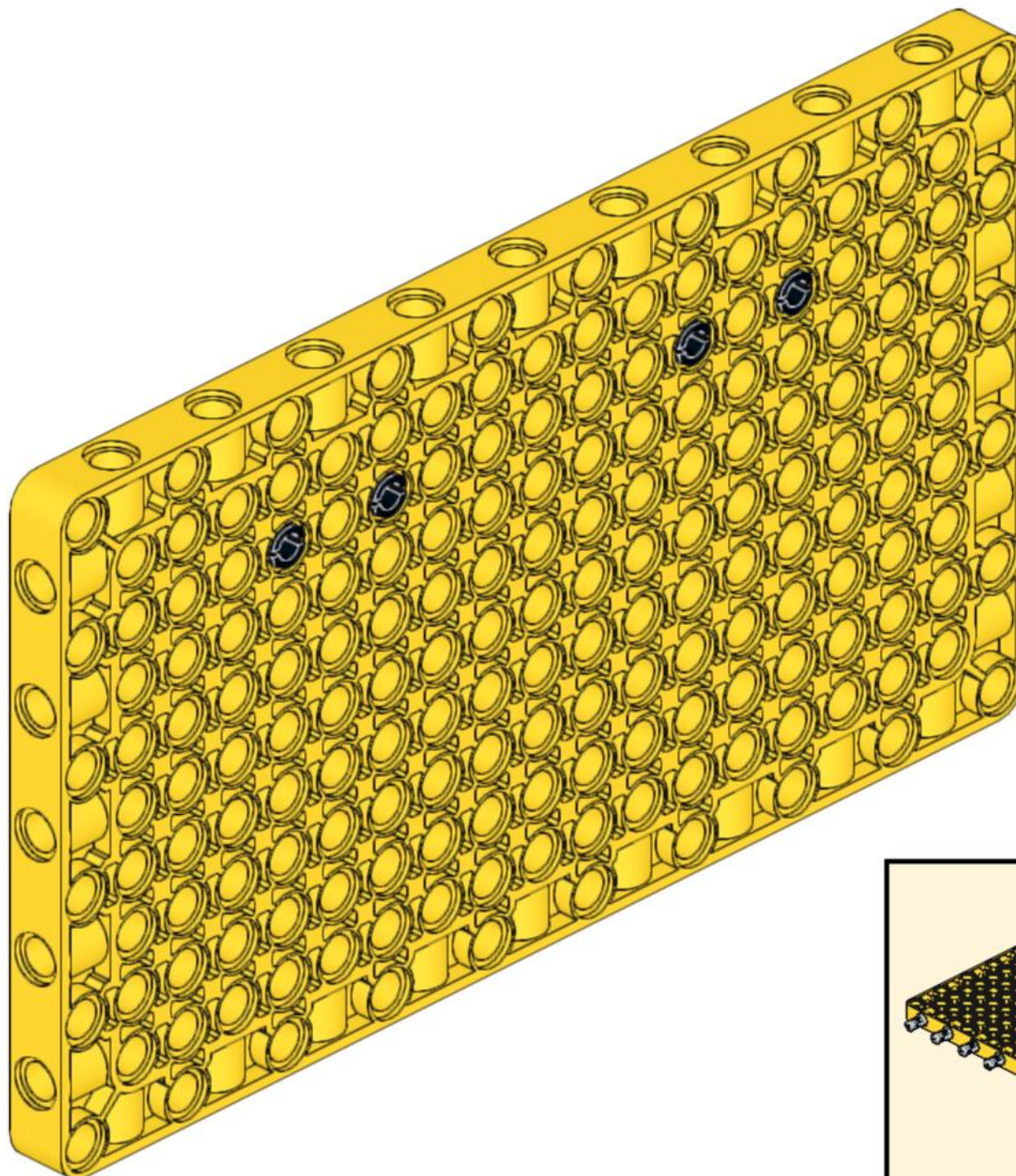
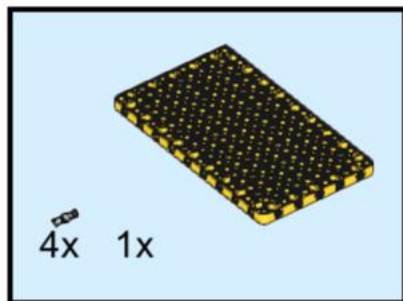
Be careful on how attachment motors are placed on top of the yellow plate covering the driving motors.

The primary mechanism the drive motors are secured is via attachment to this plate.

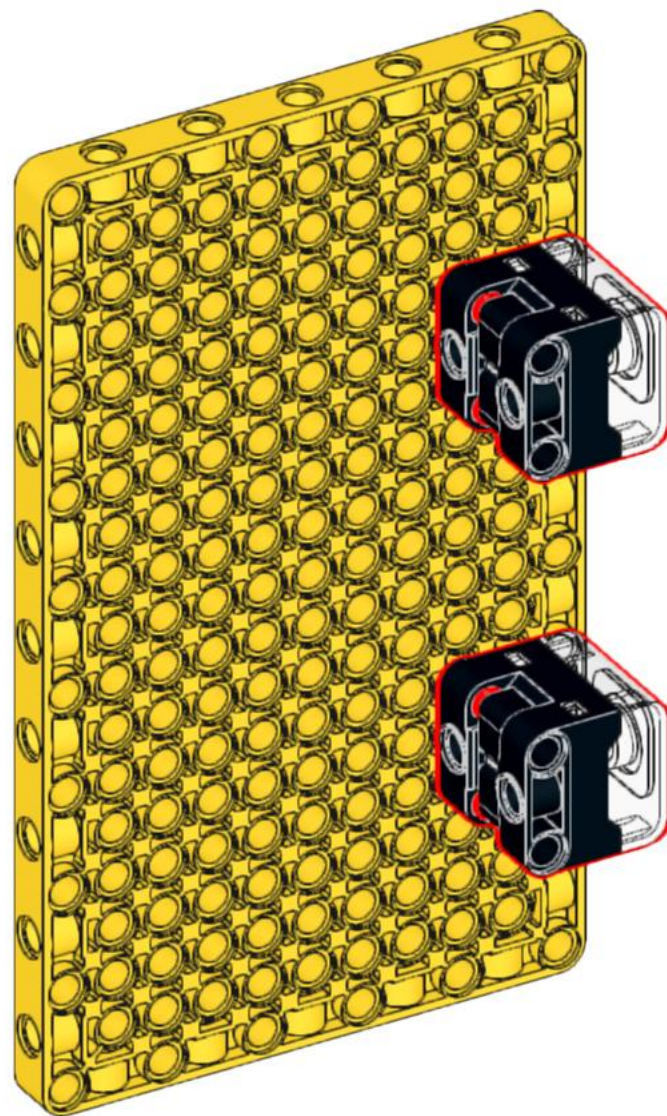
You need to optimize the number of pins (8) from the drive and attachment motors that are connected to the top and bottom of the plates.



1

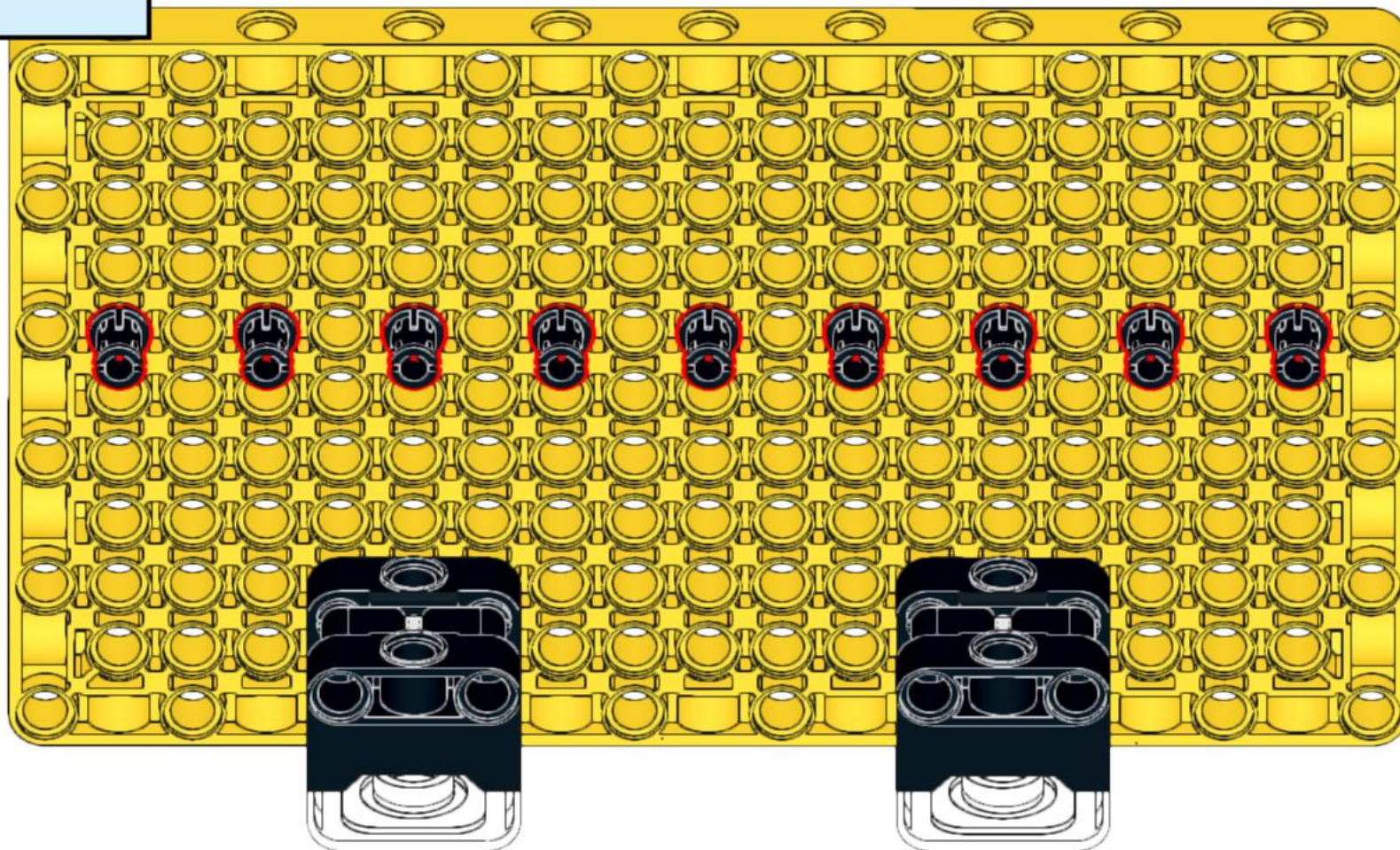


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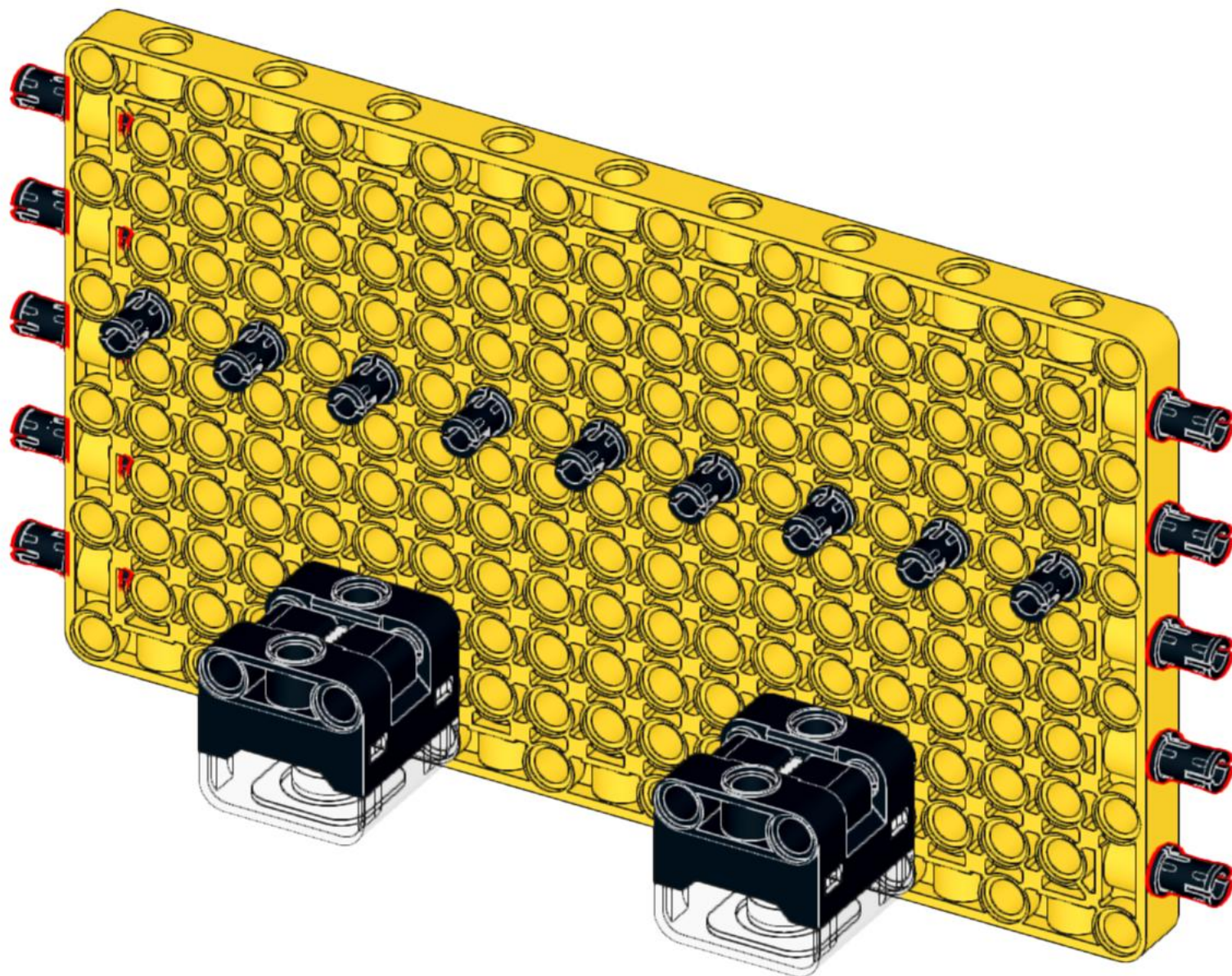


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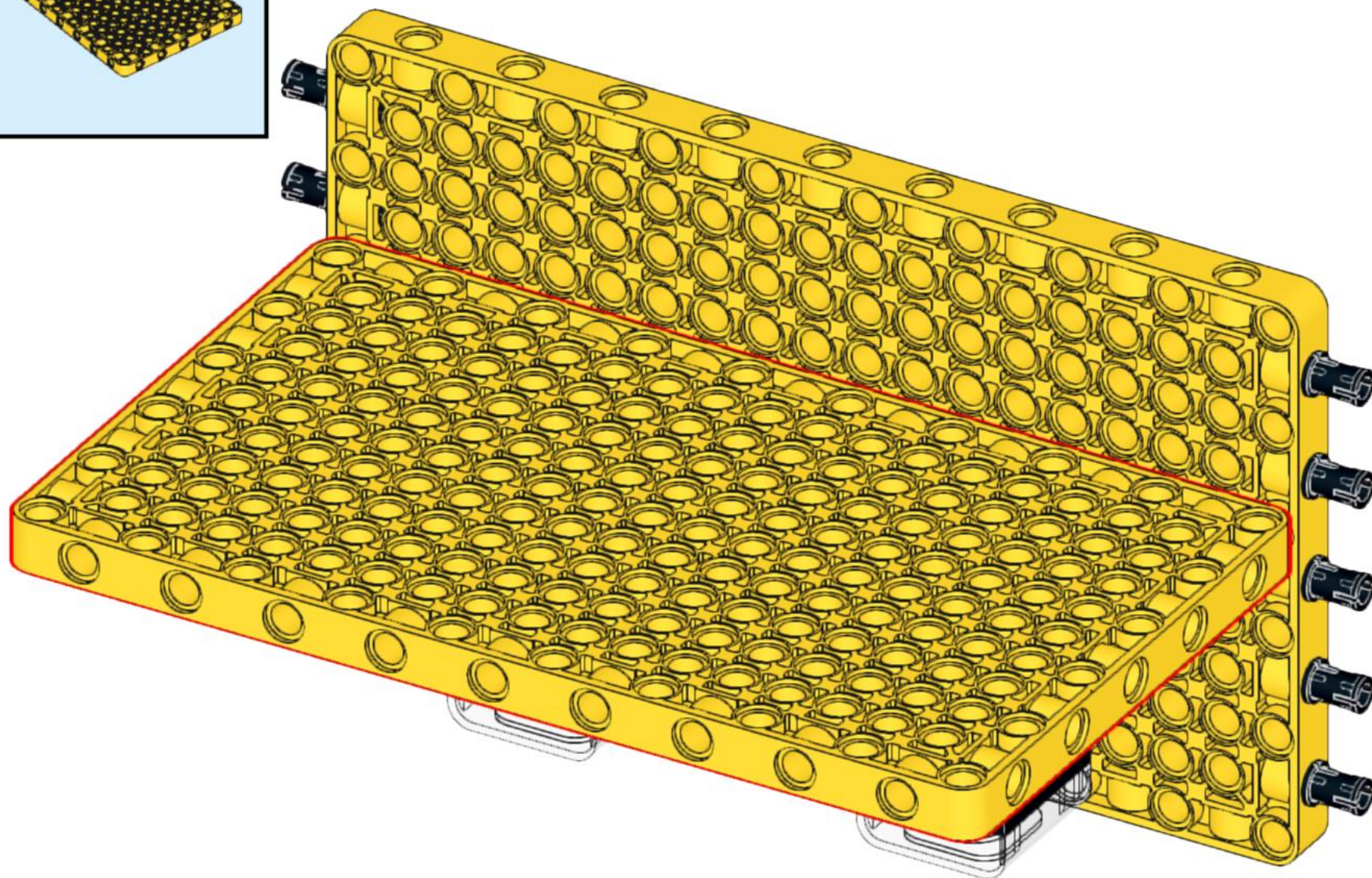
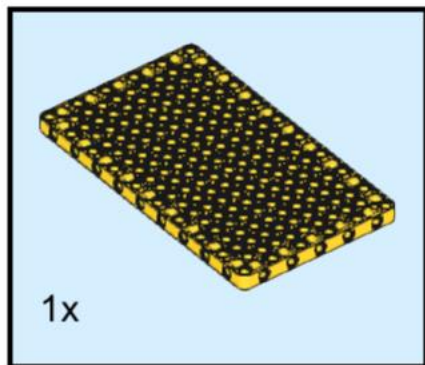


4



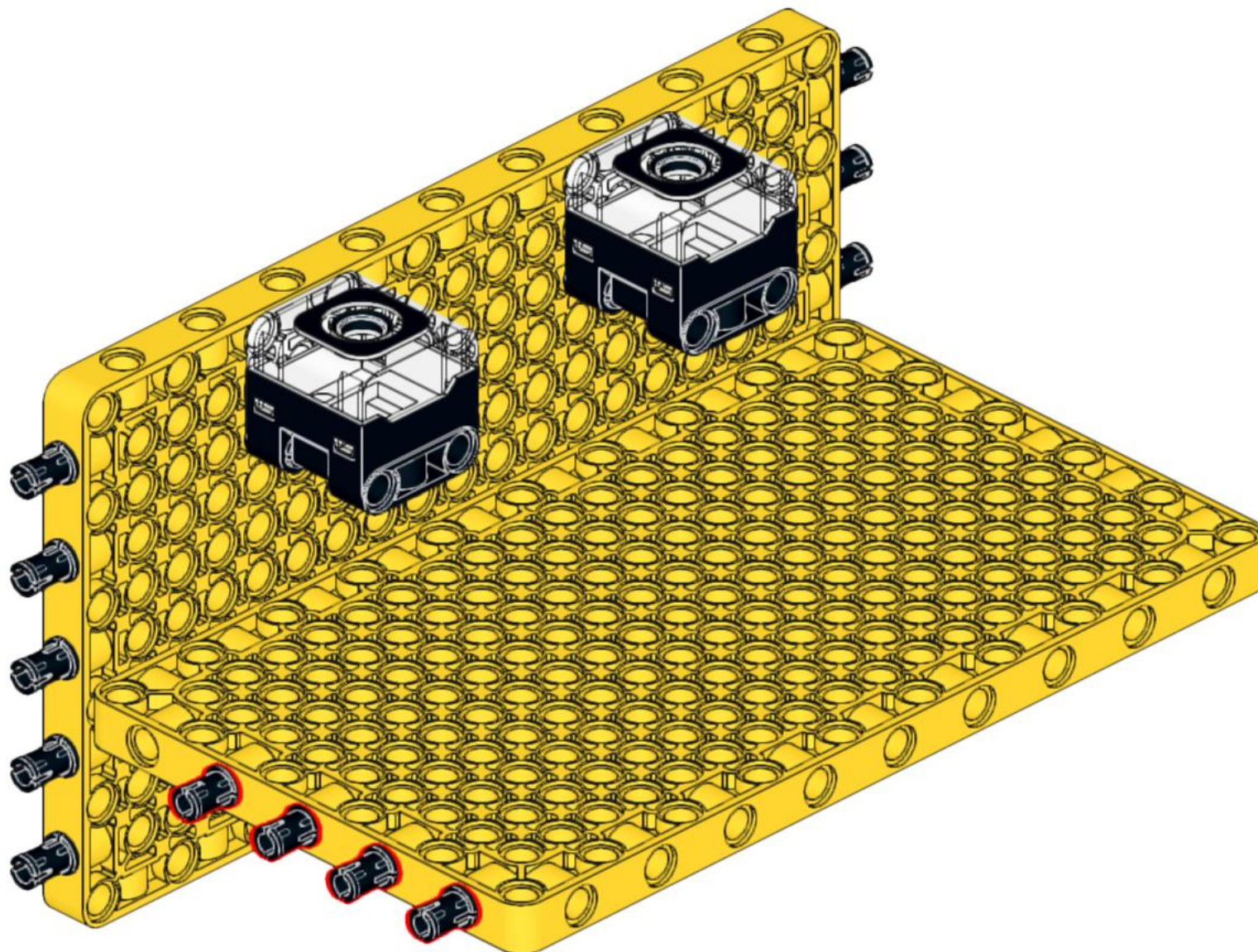


5



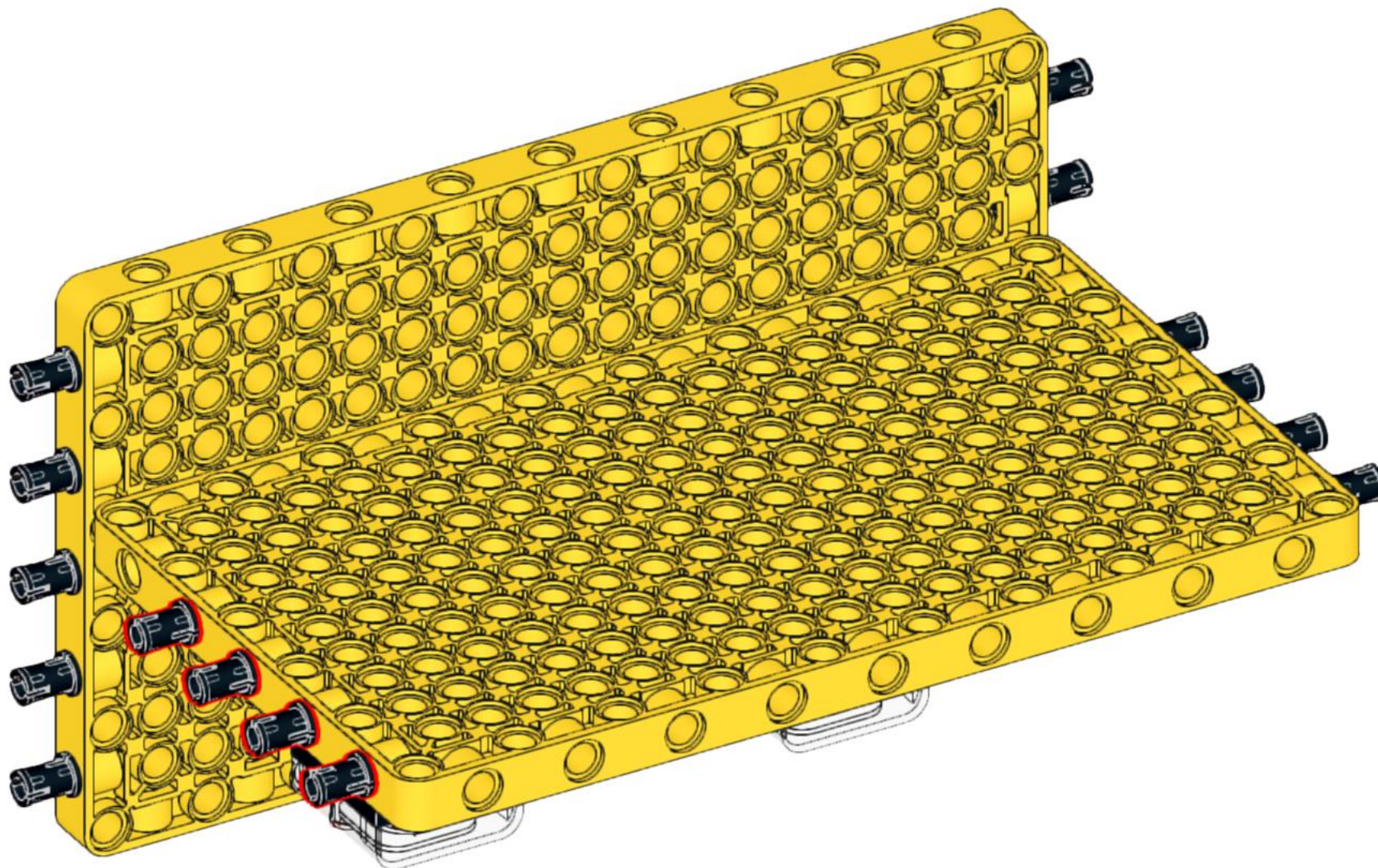


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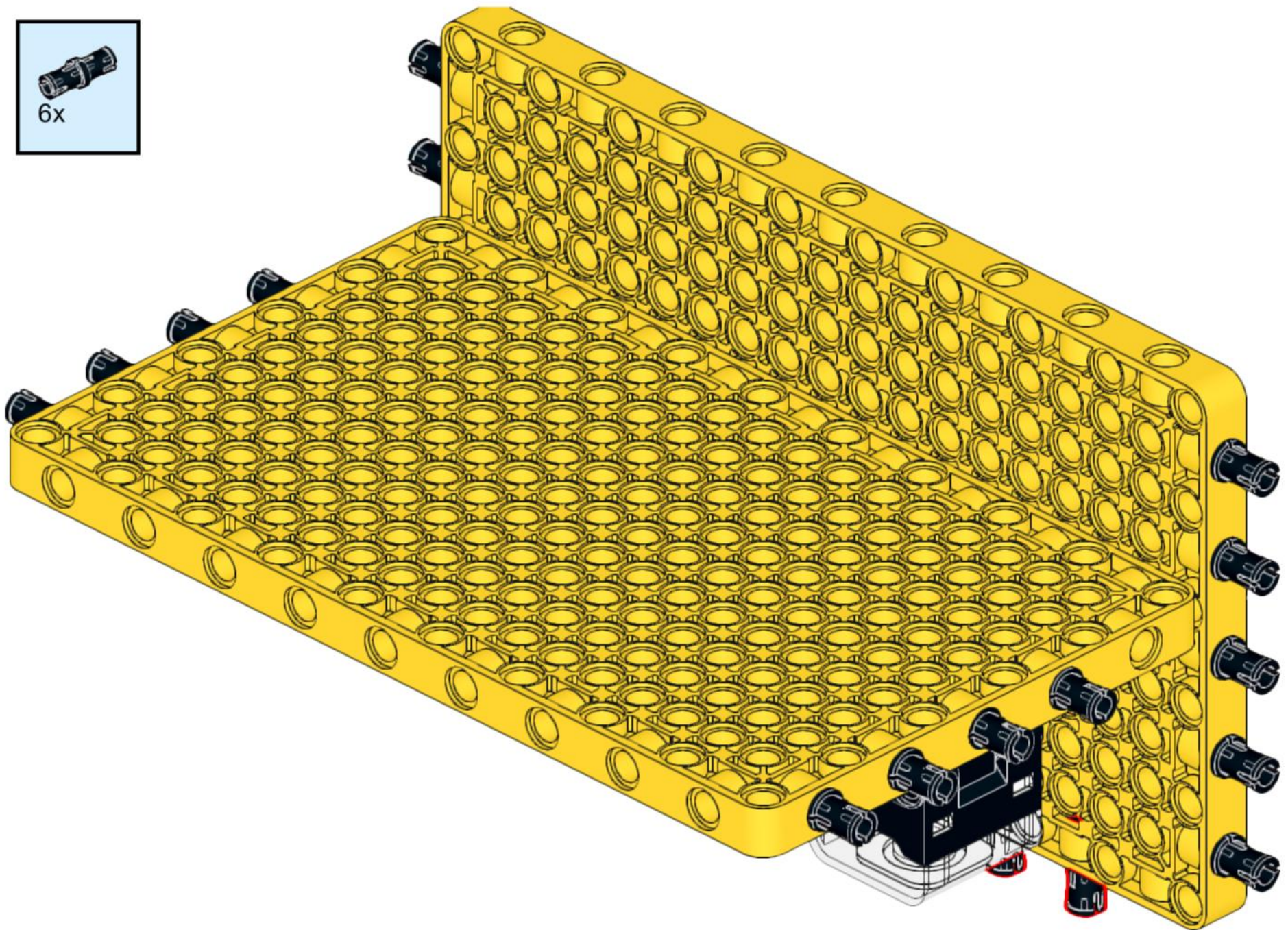


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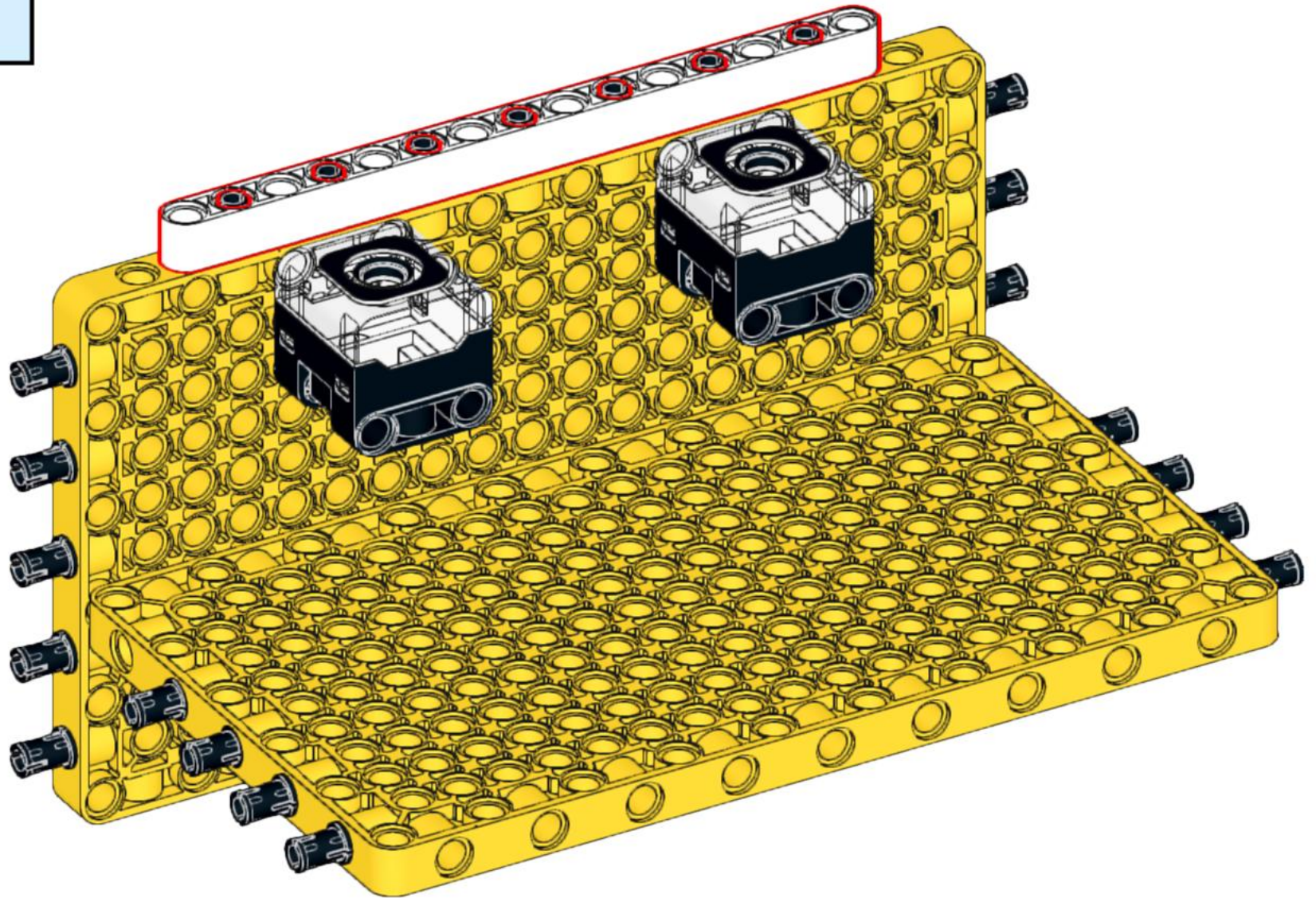
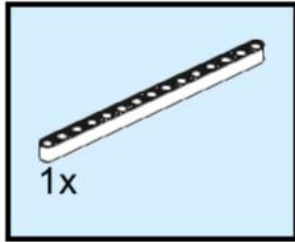


8

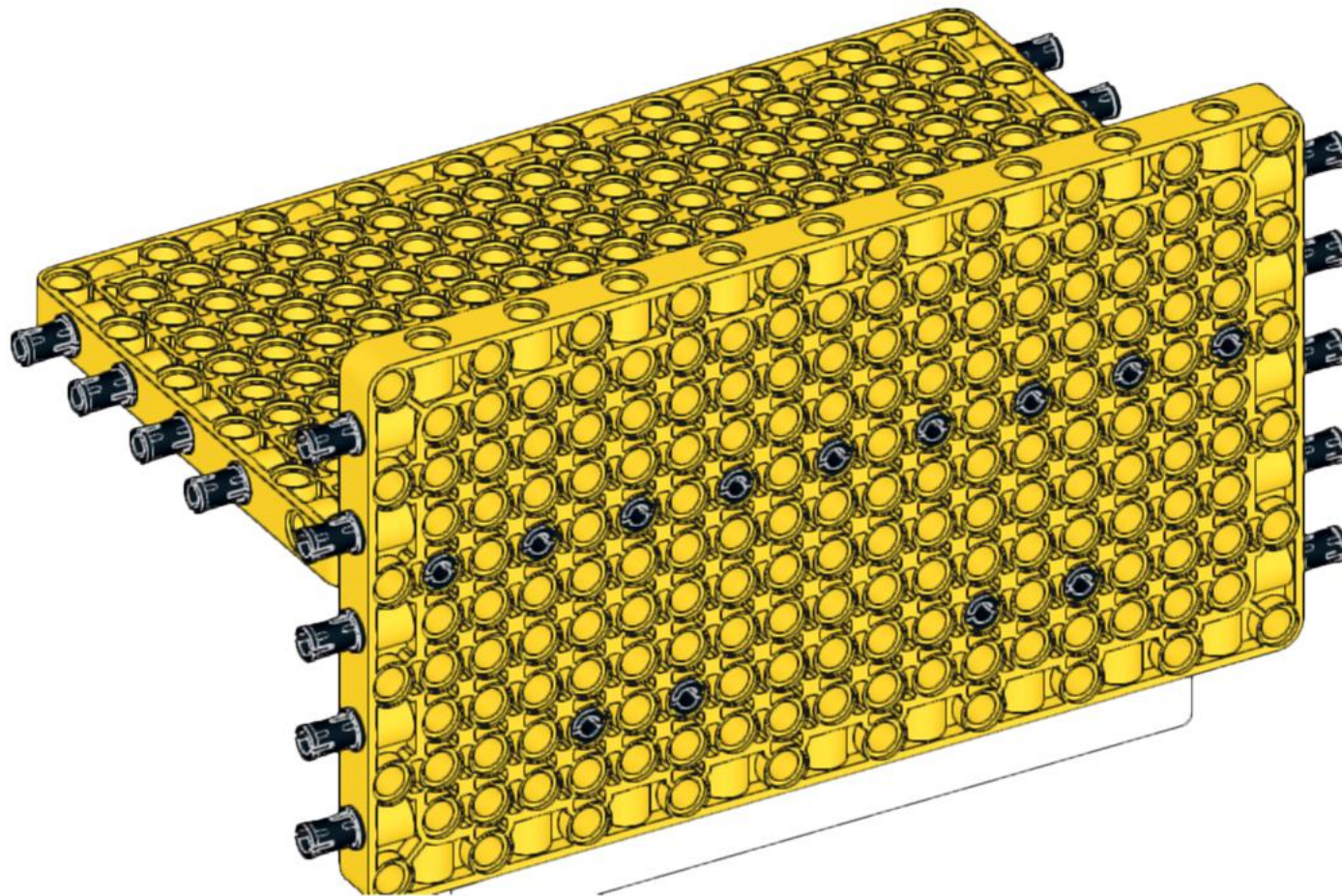




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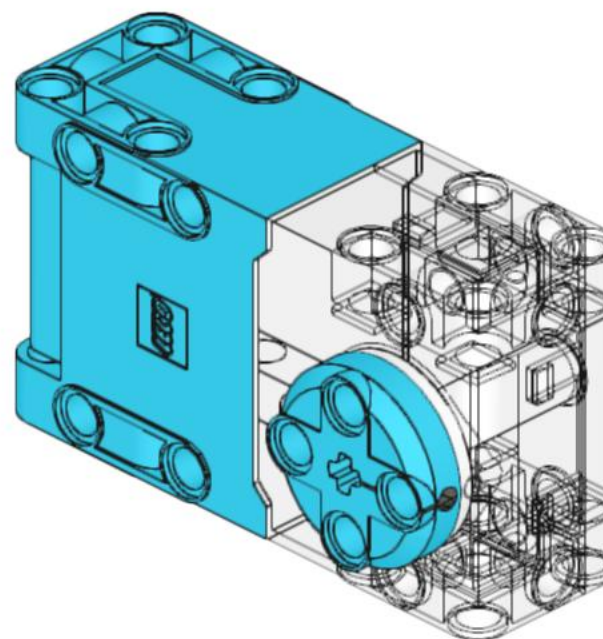
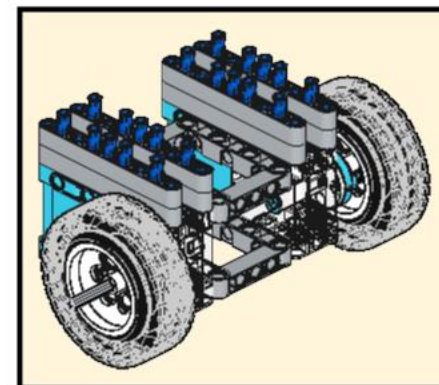
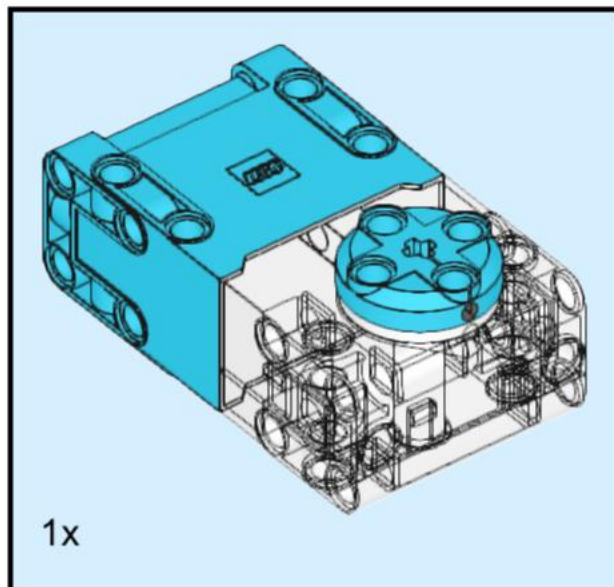


# 10

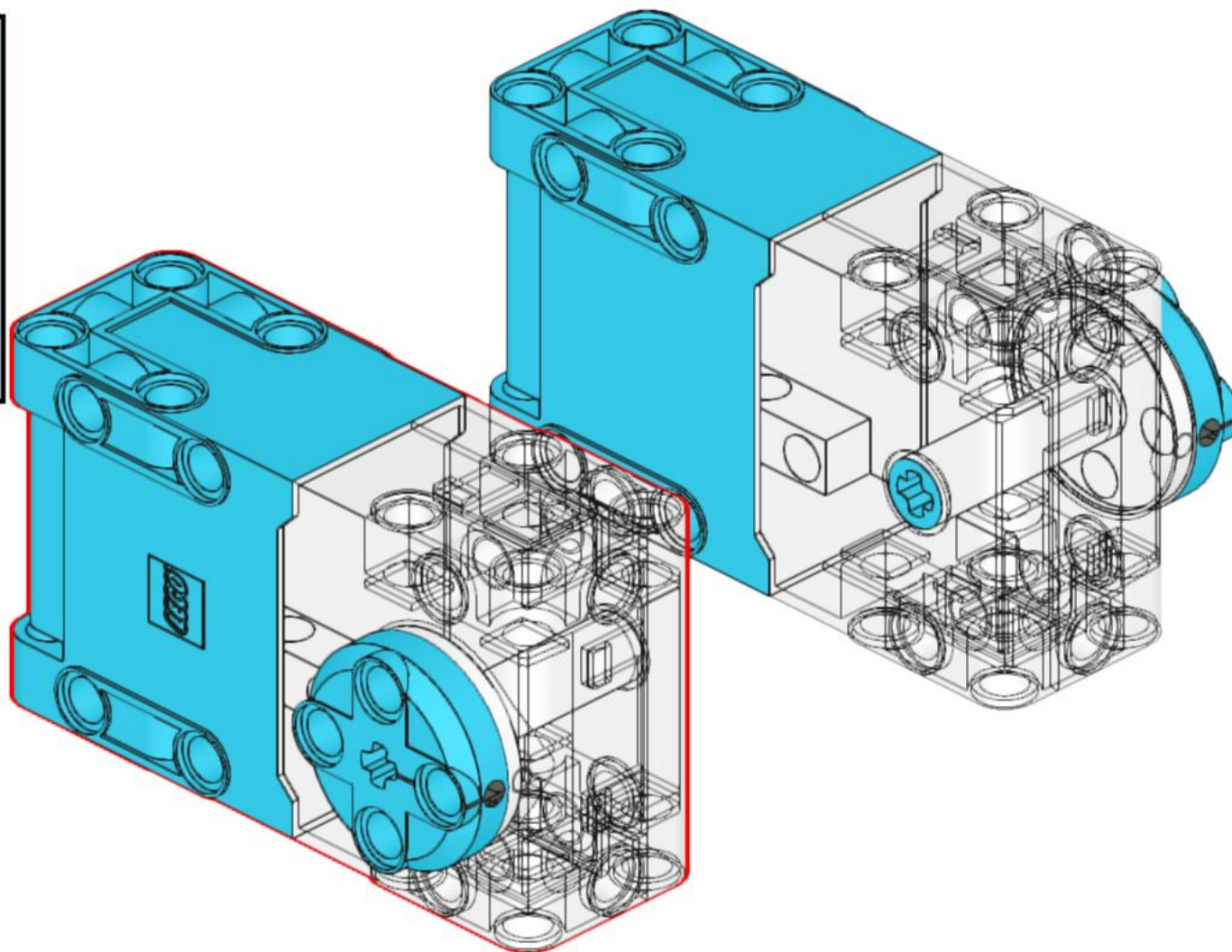
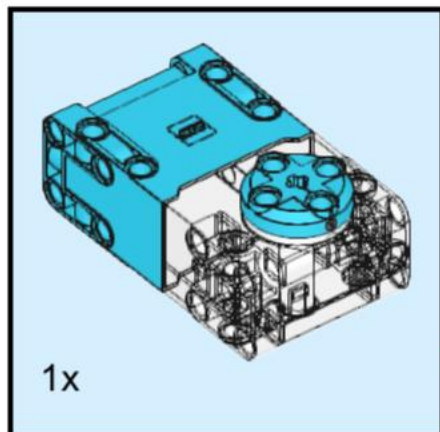




# 11

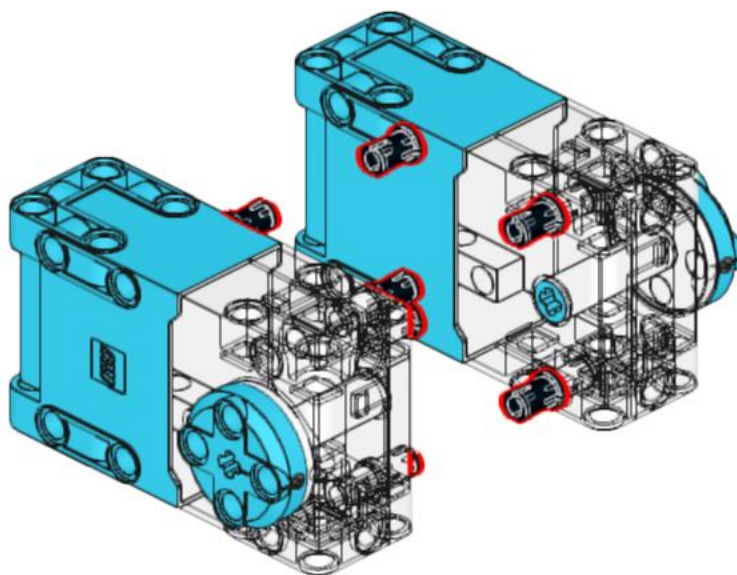


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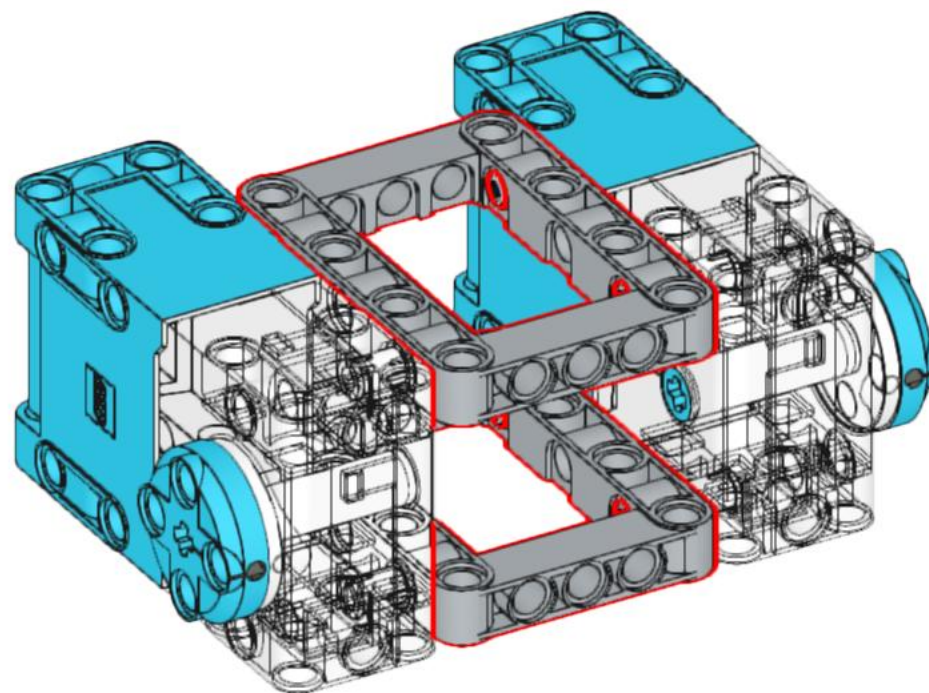
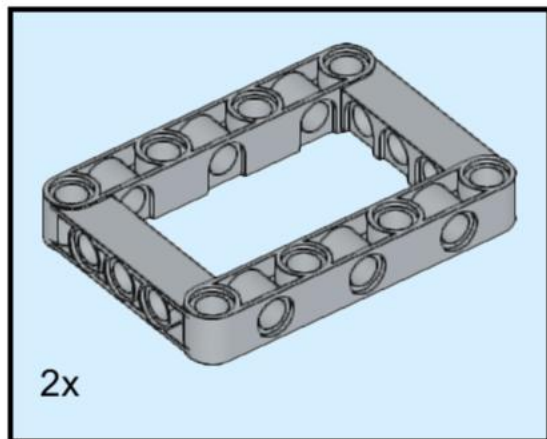




# 13

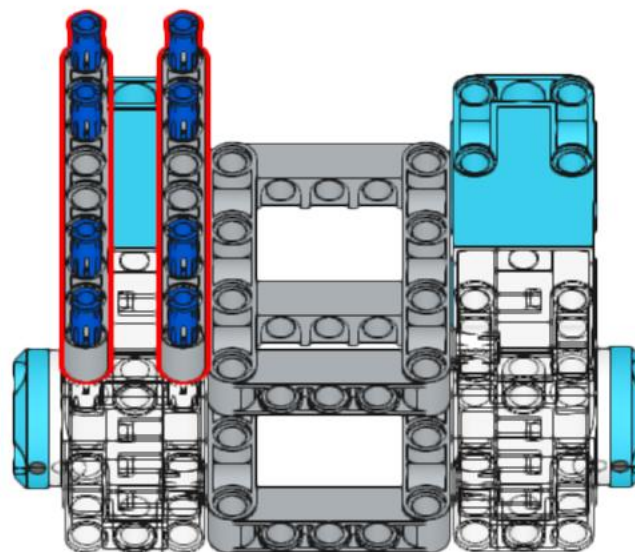
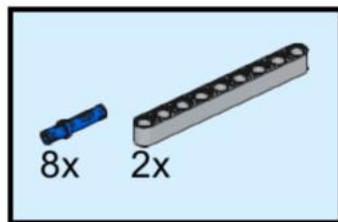


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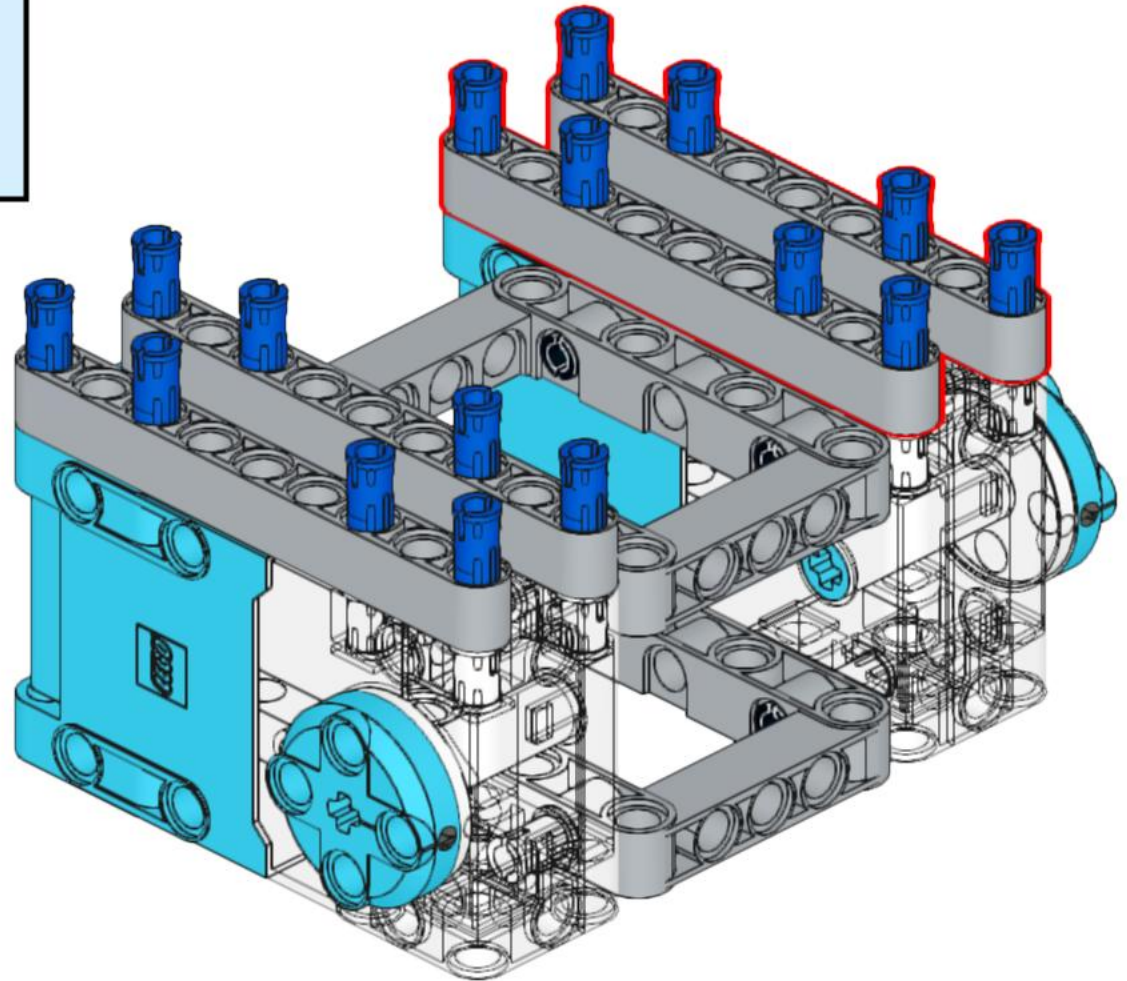
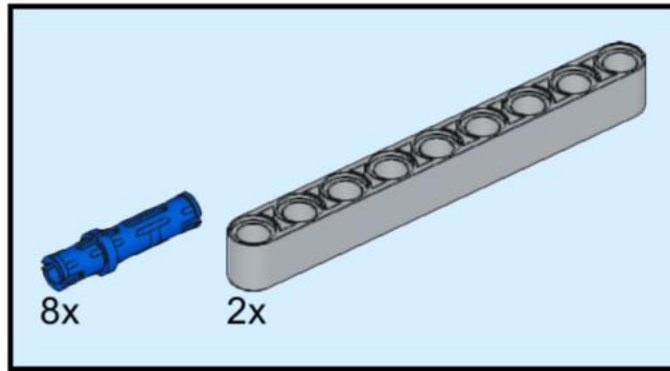




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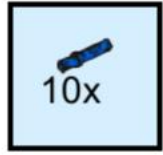


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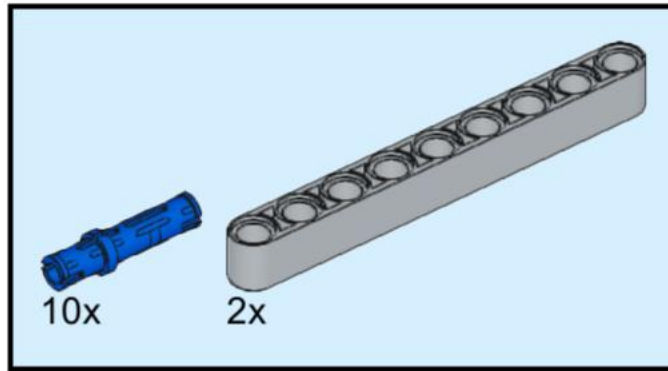




17



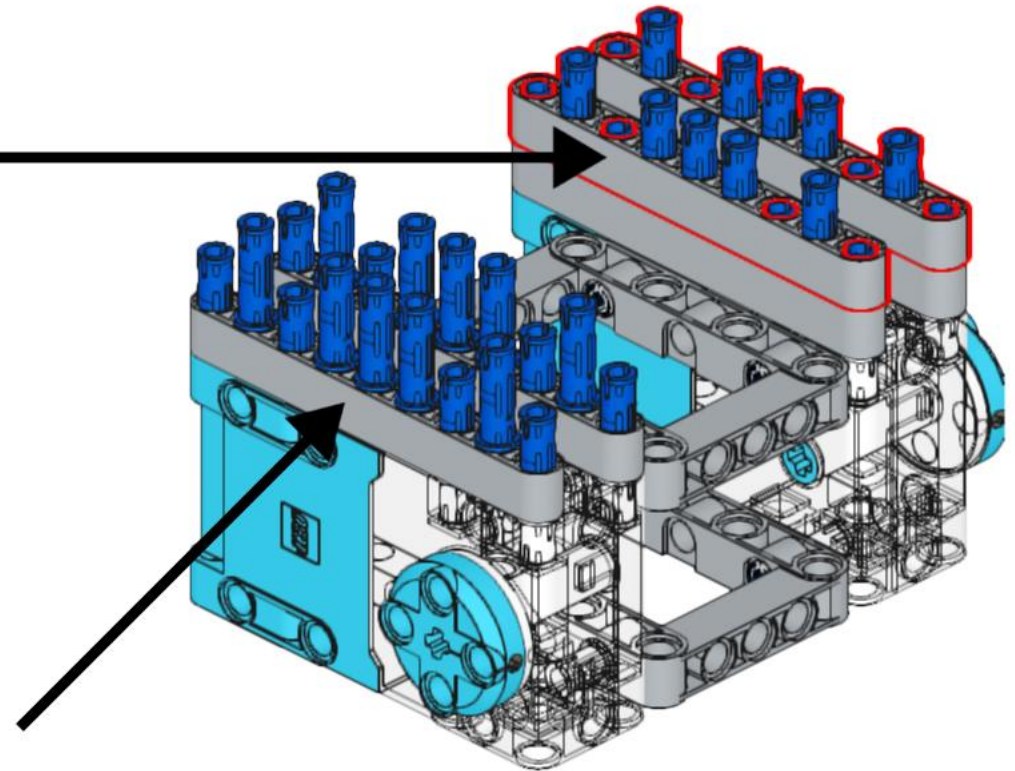
# 18



You might need to change which 3L pins are present depending on how your left and right attachments align up on the 11x19 plate that sits on top of the drive motors.

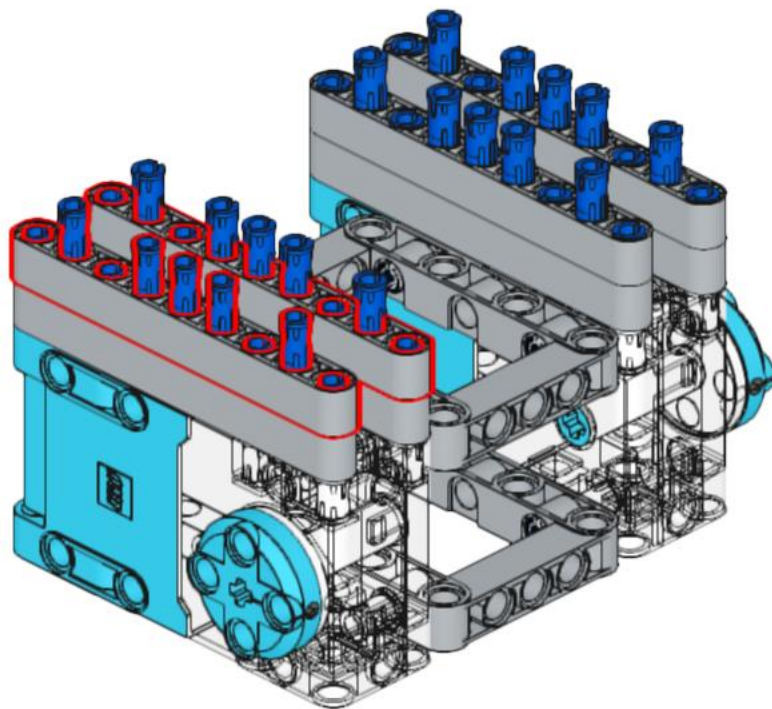
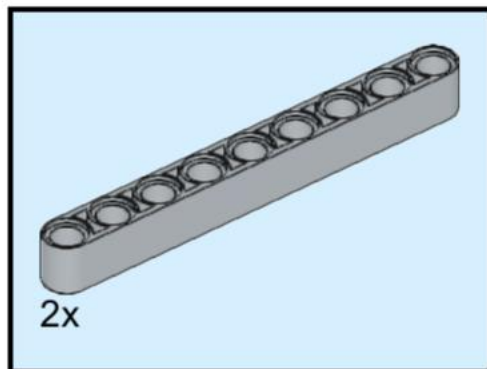
Another option is instead of recessing all the 3L Pins into the bottom gray beam, you can instead connect them to the top gray beam so they will penetrate the 11x19 plate and connect to your attachment motors.

It is a tradeoff in strength vs. flexibility.

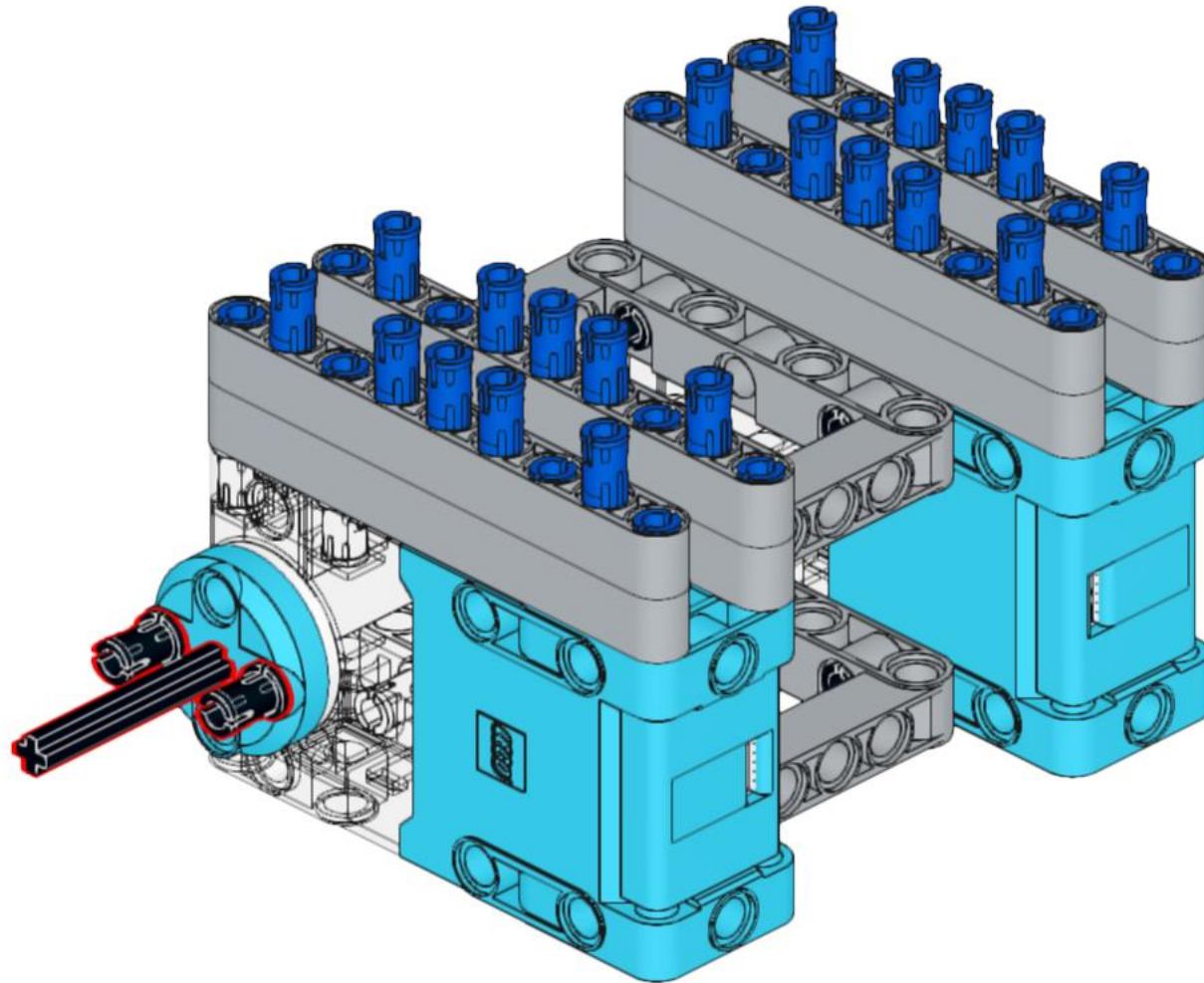
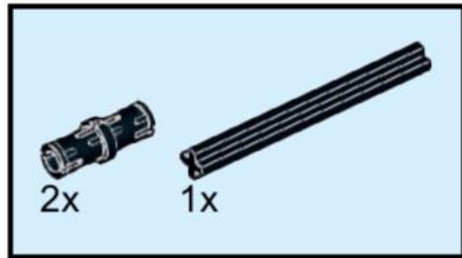




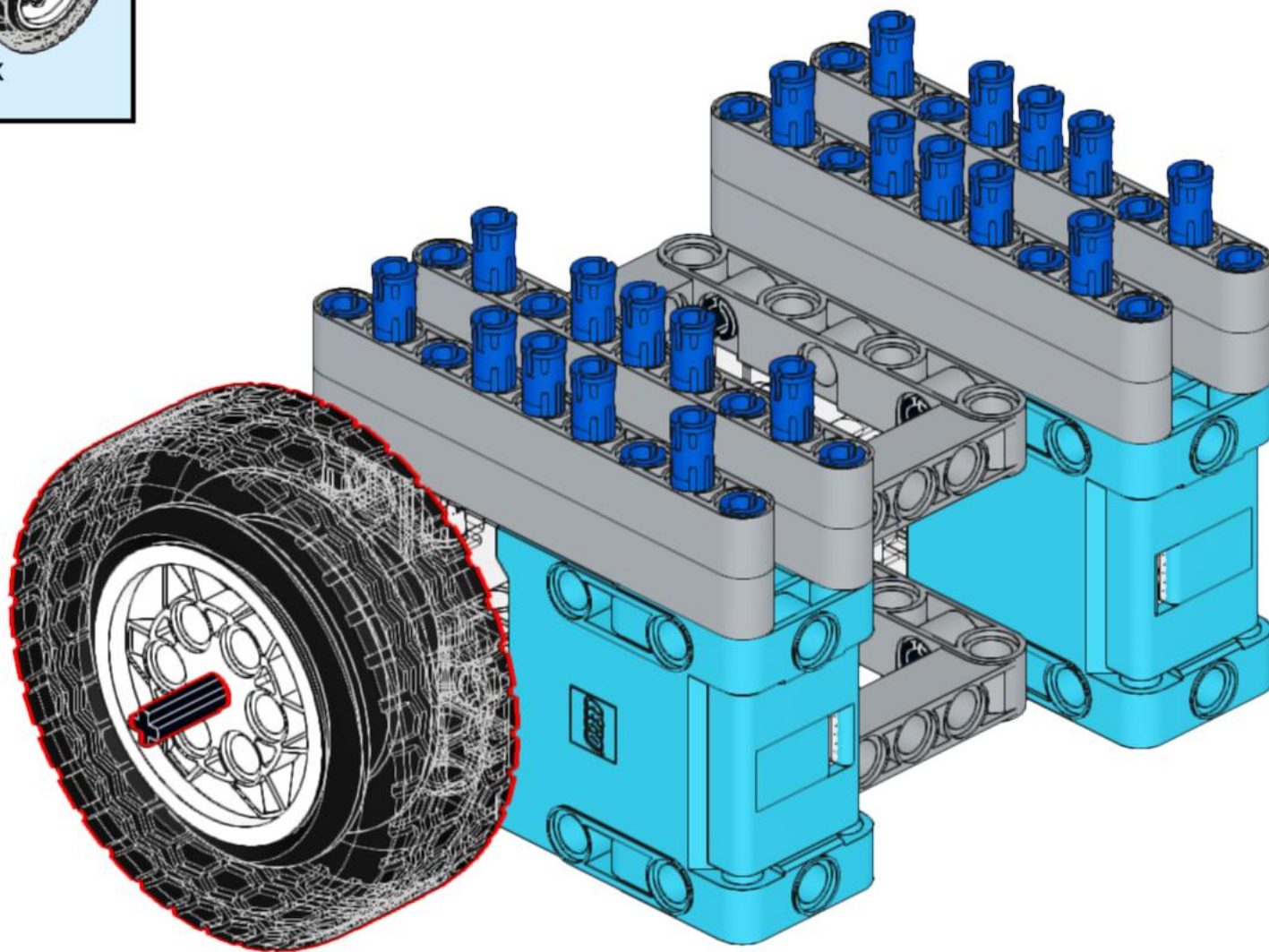
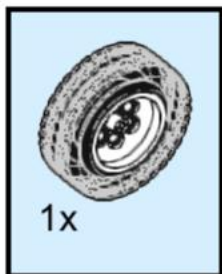
# 19



# 20

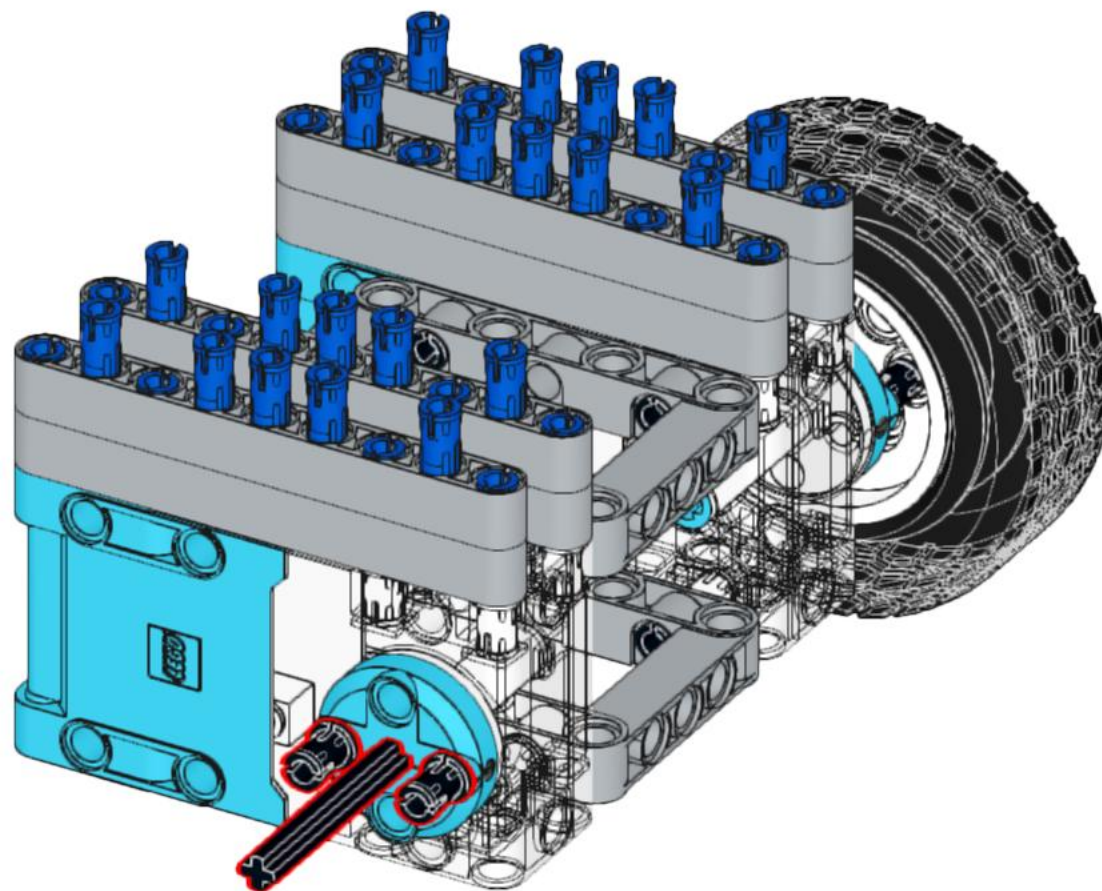
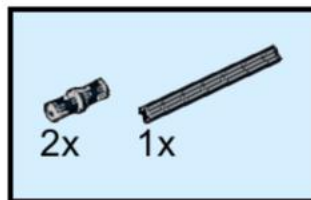


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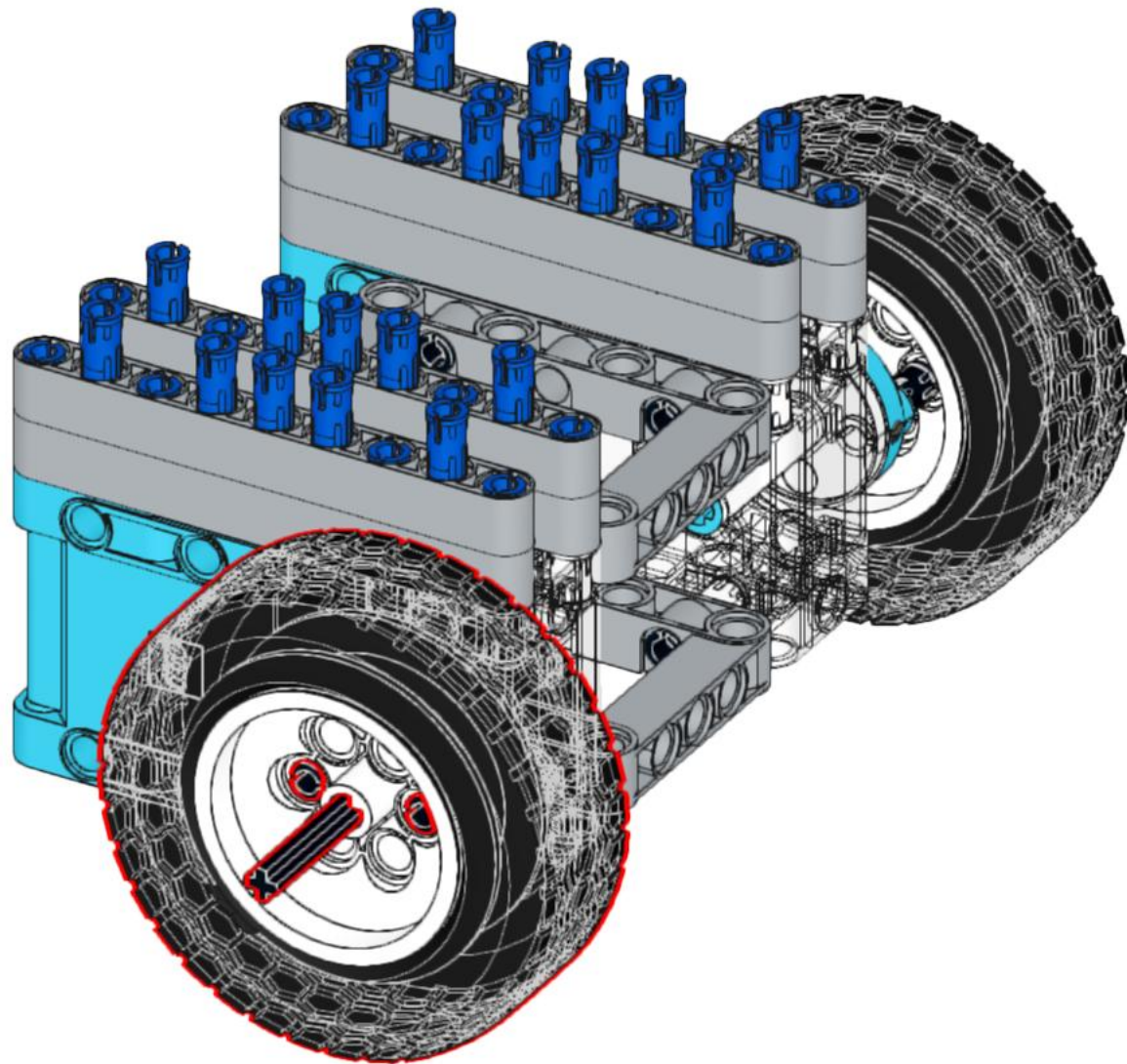
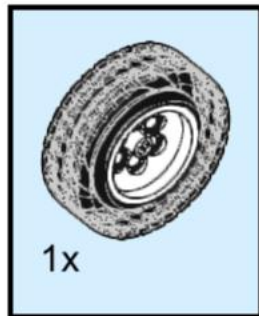




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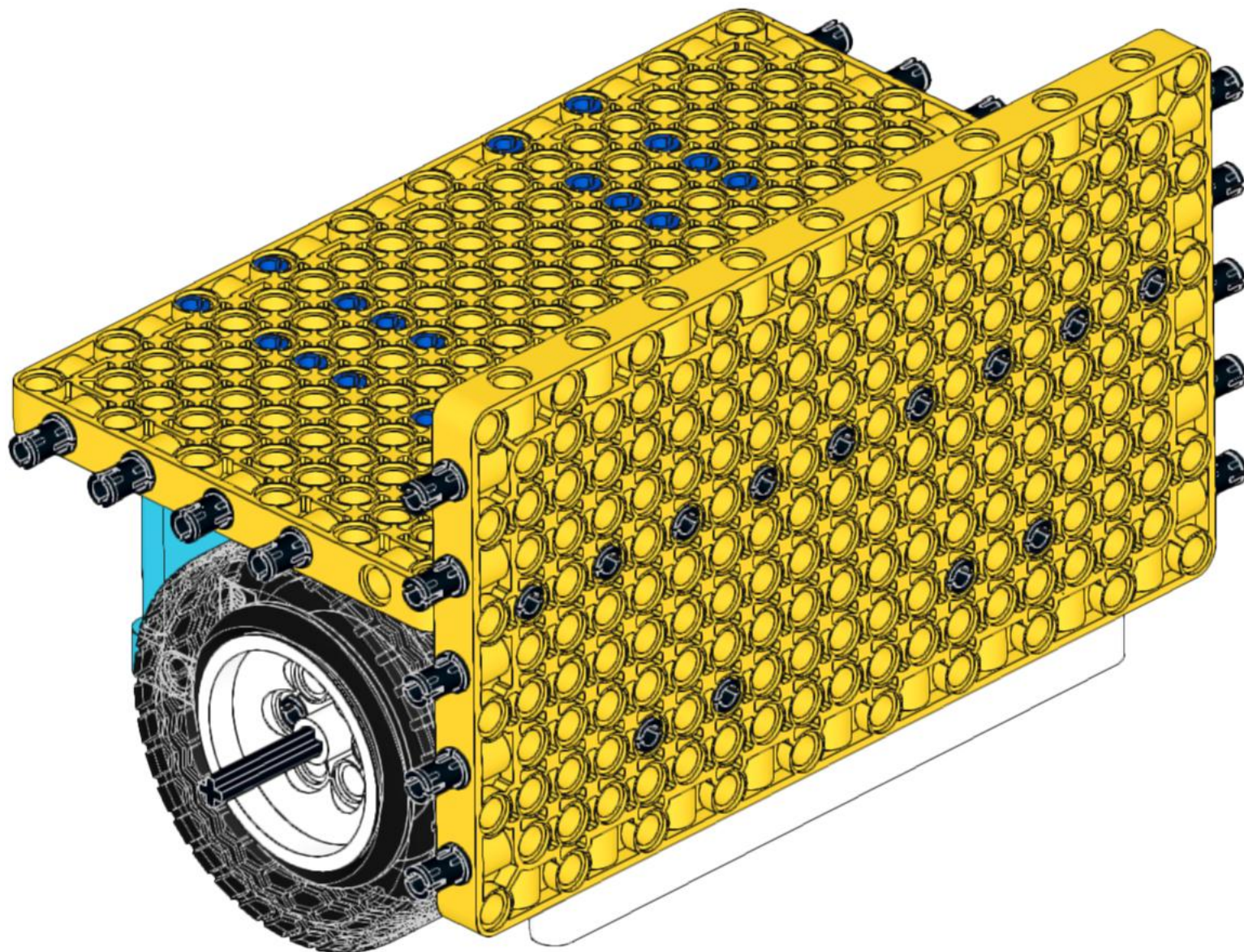


# 23





24



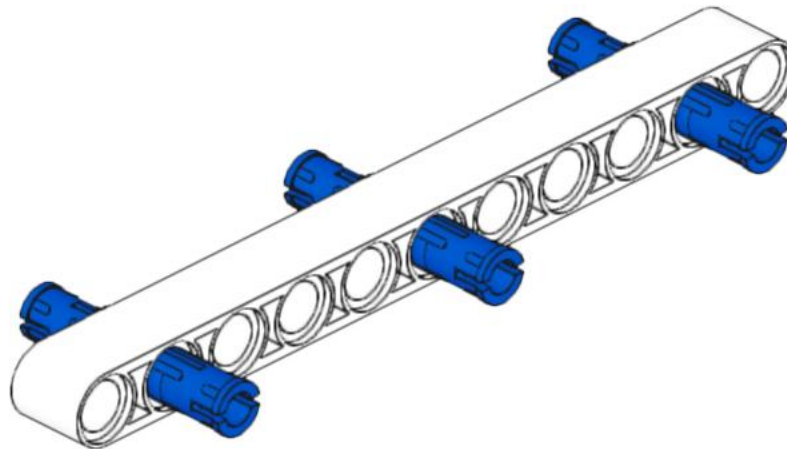
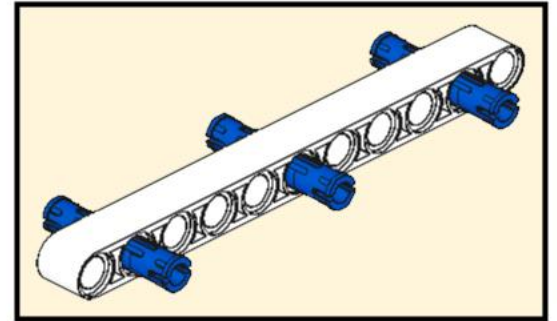
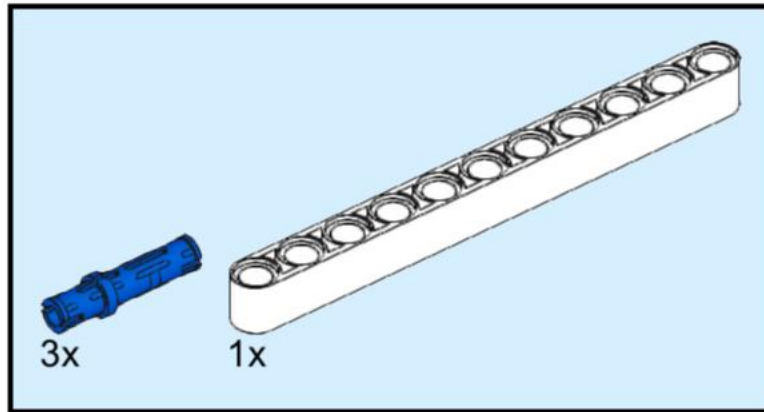


## Rear Assembly

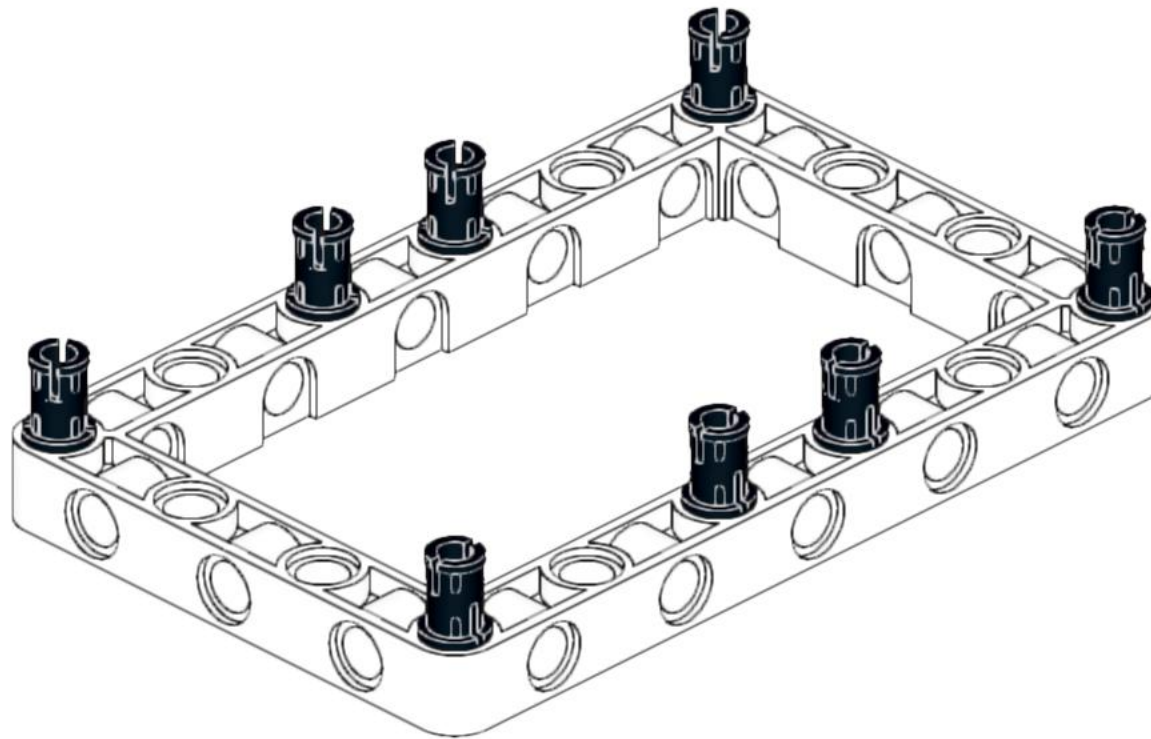
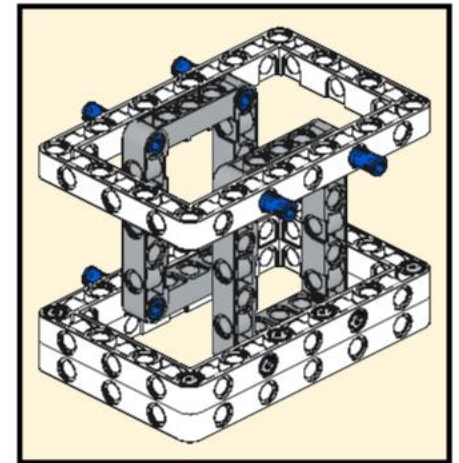
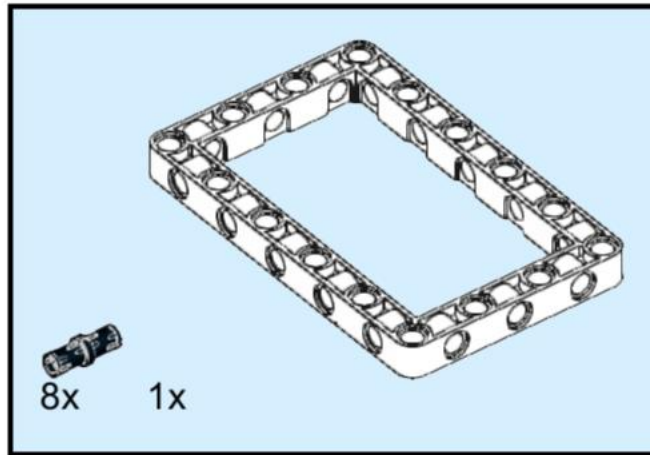
This is the portion of the robot that contains the hub.

While not shown in these plans, we sometimes installed two 2×6×2 Brick, Counter Weights (73090) in the middle 5x7 frame to offset the weight of the front of the robot.

# 25

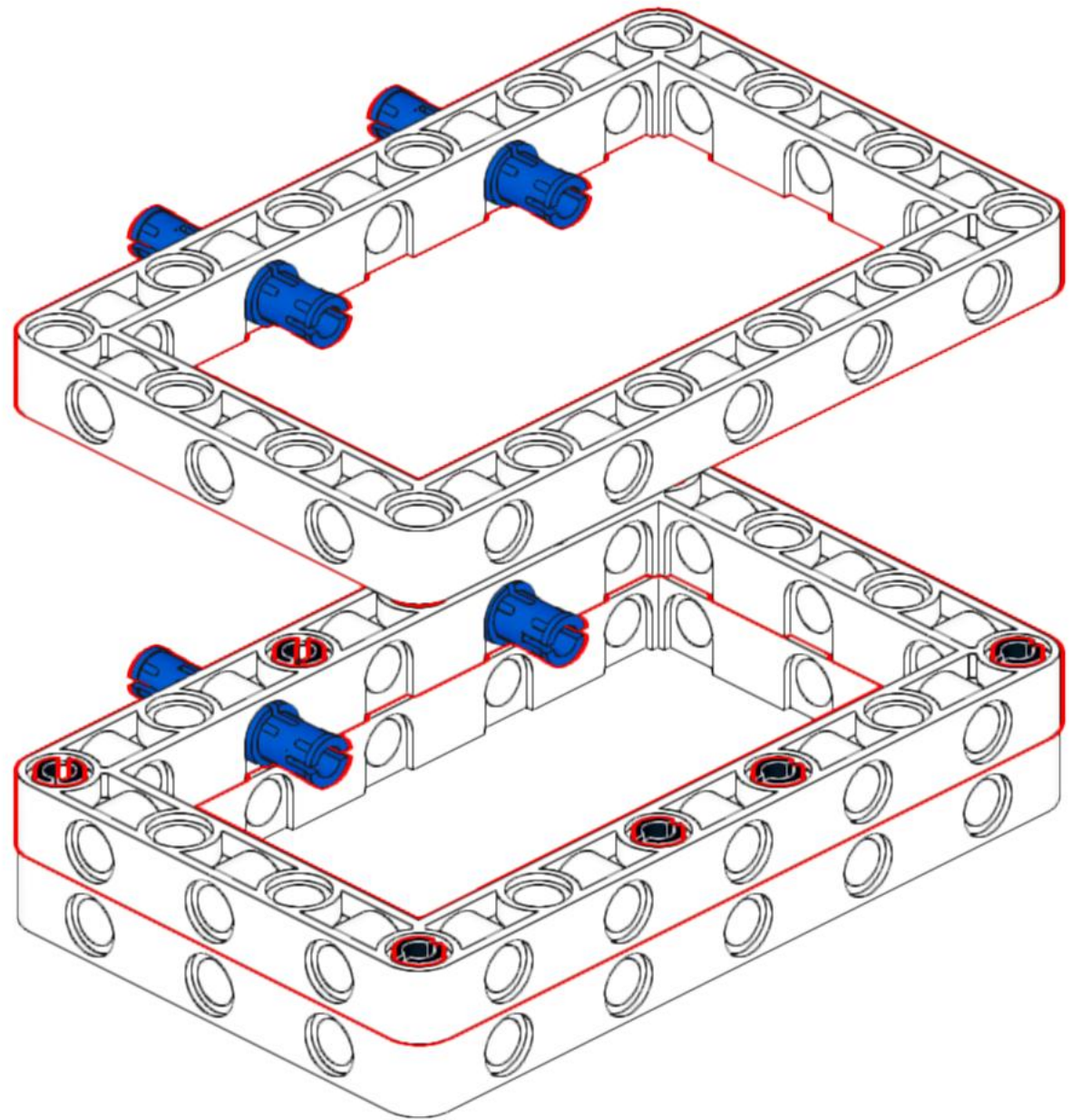
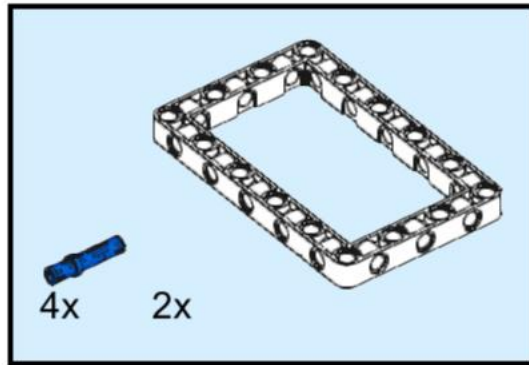


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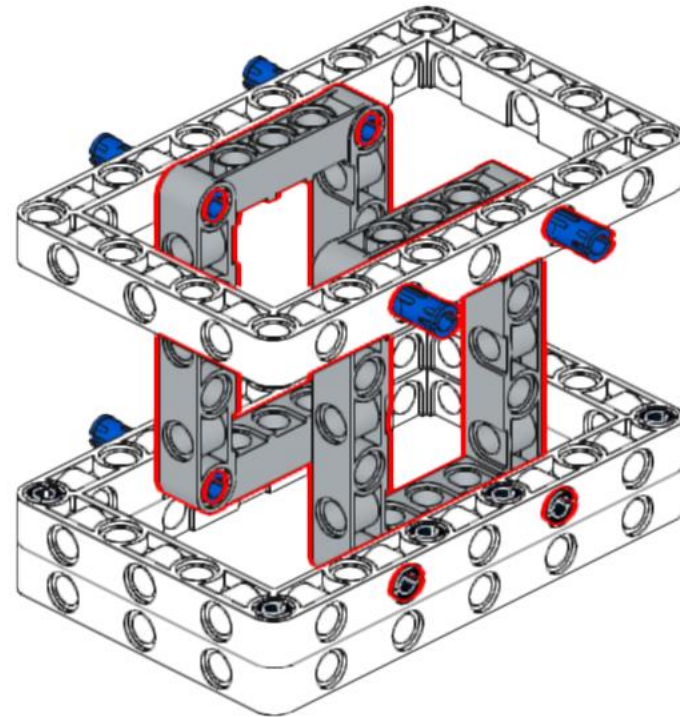
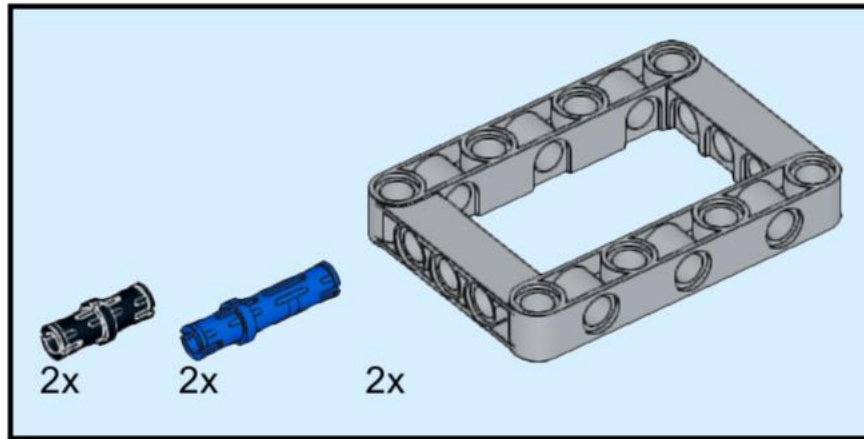




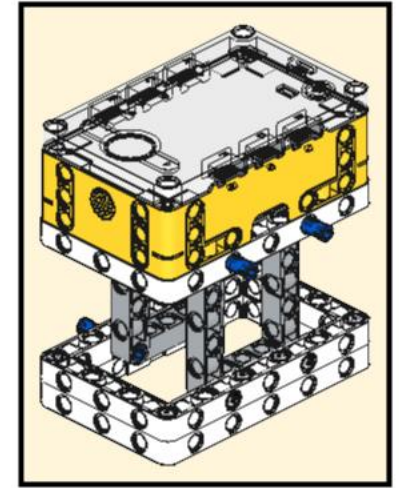
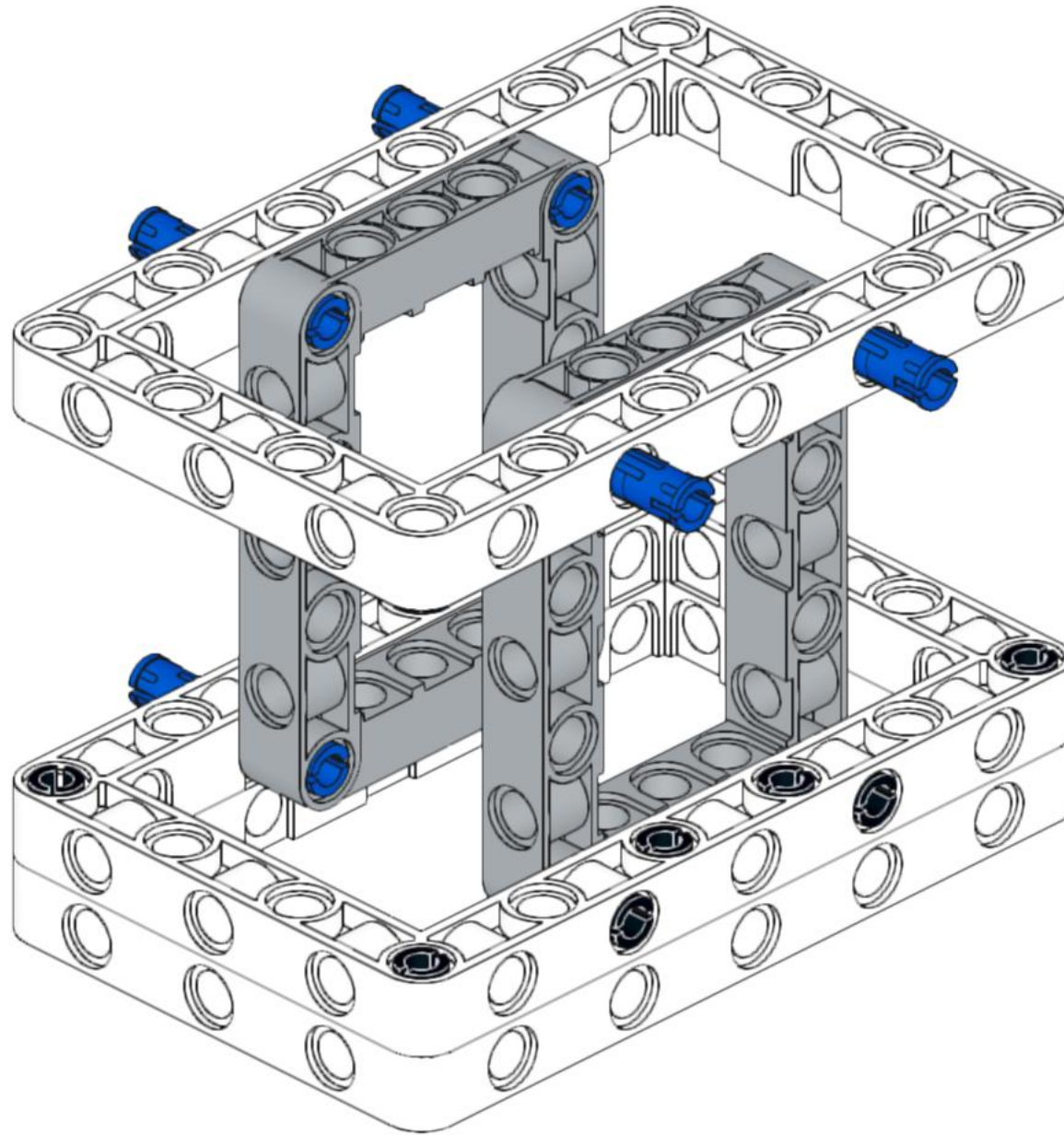
# 27



# 28

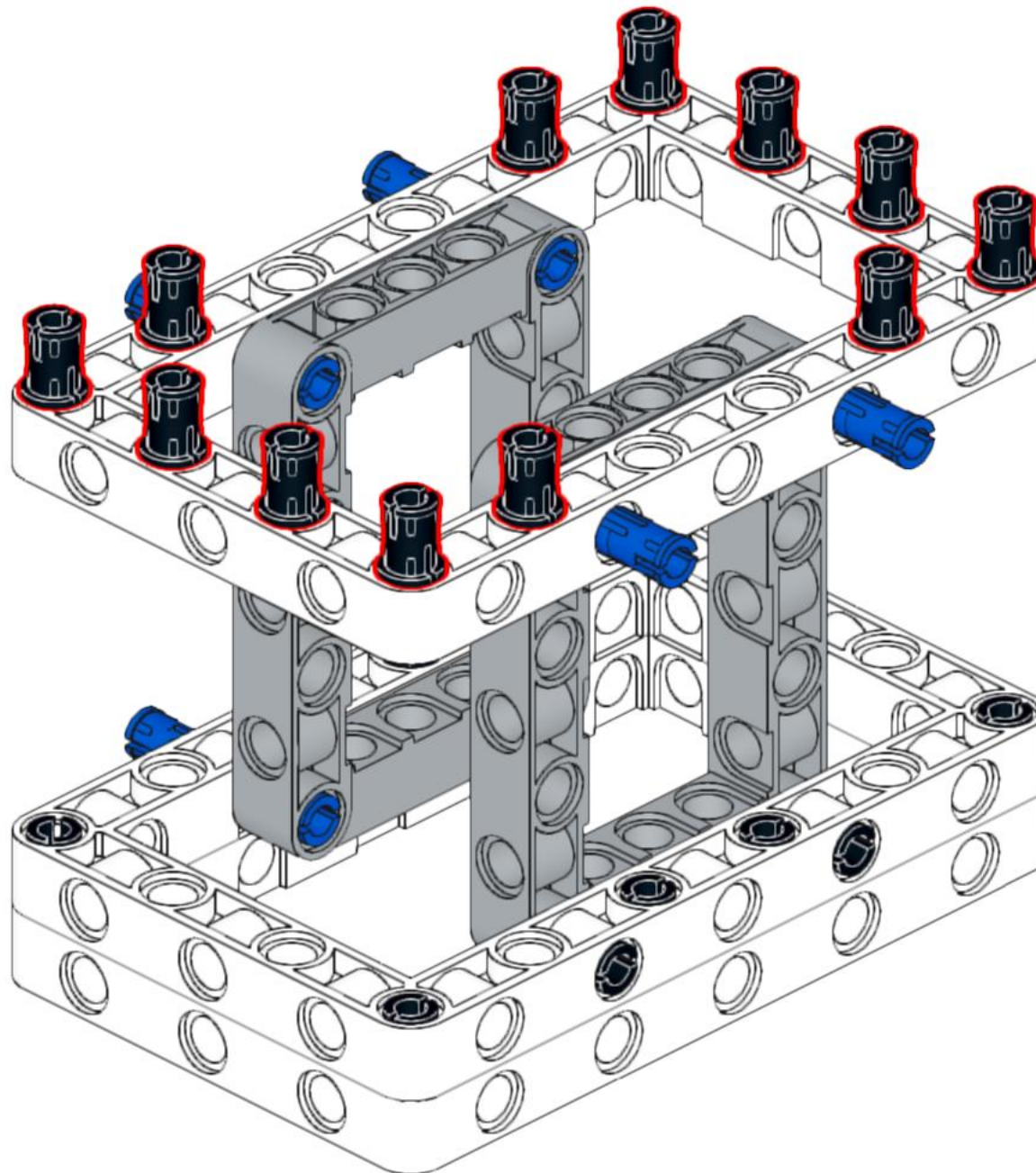


# 29

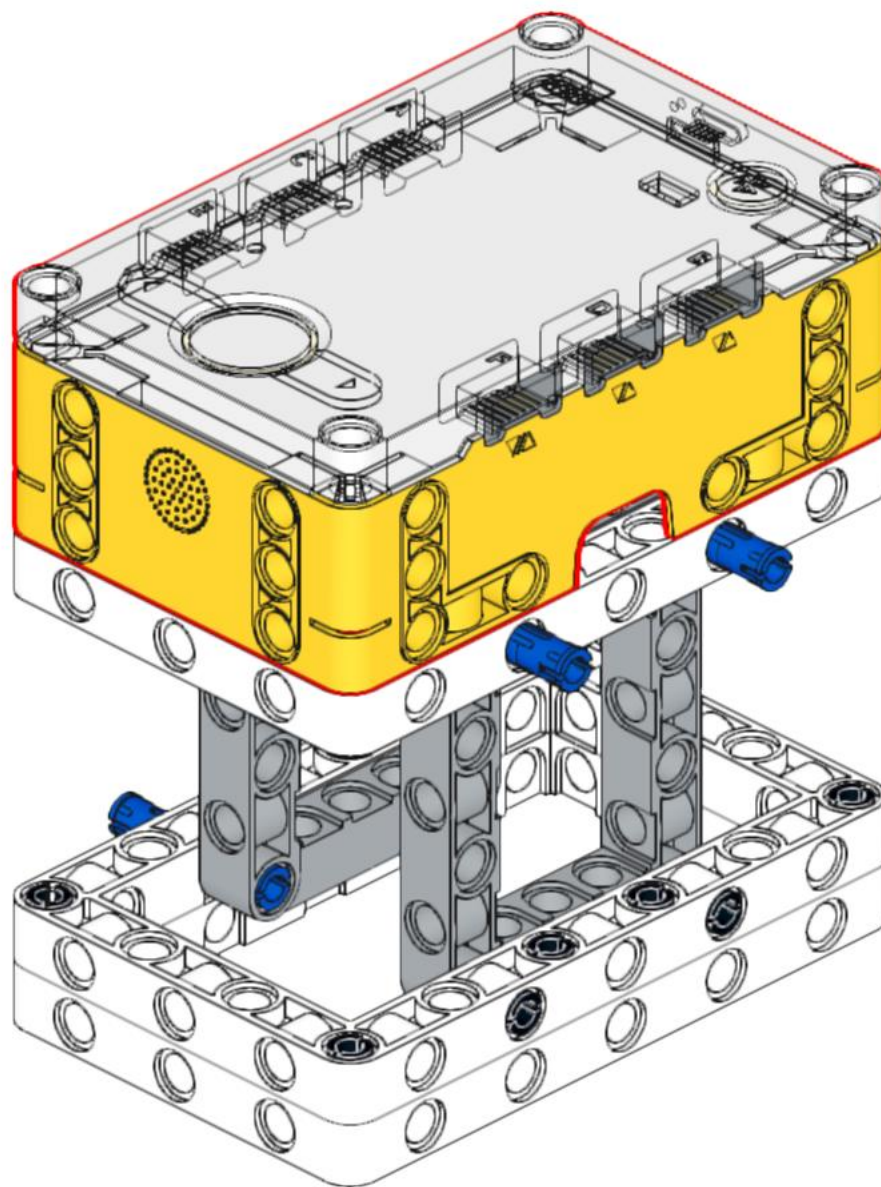
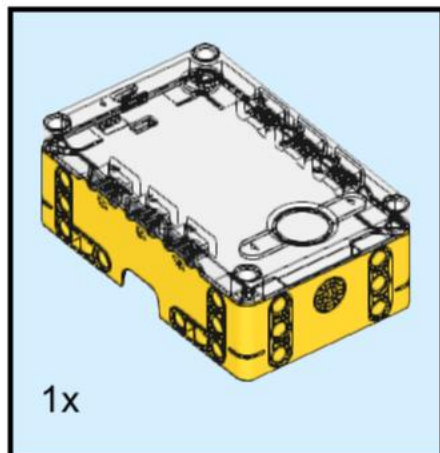




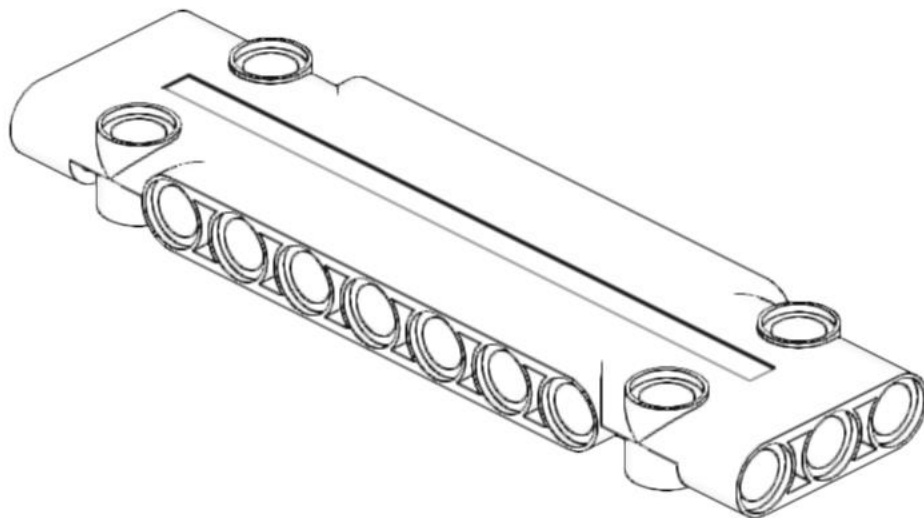
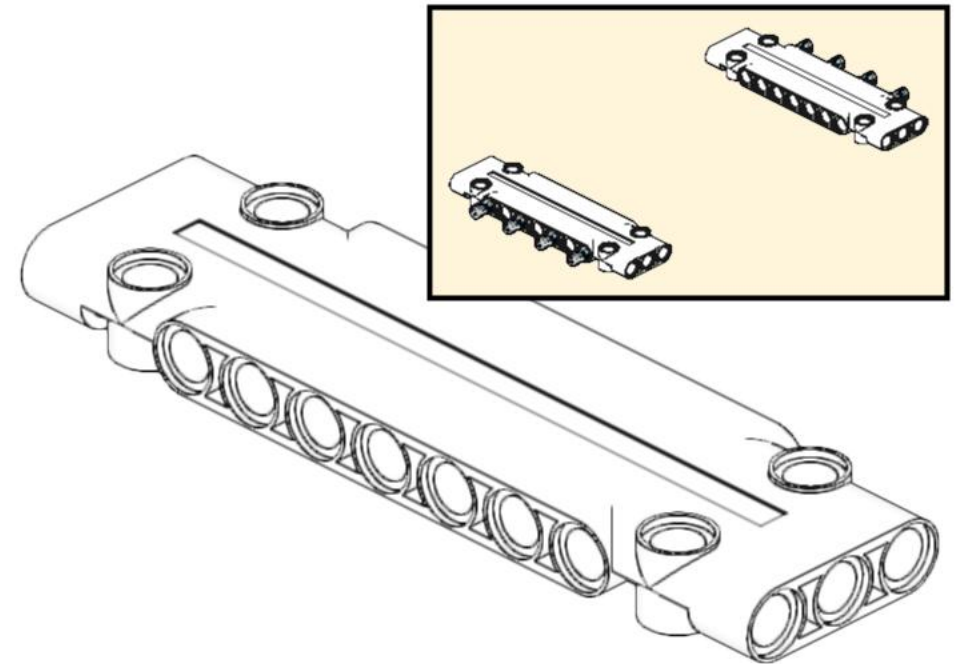
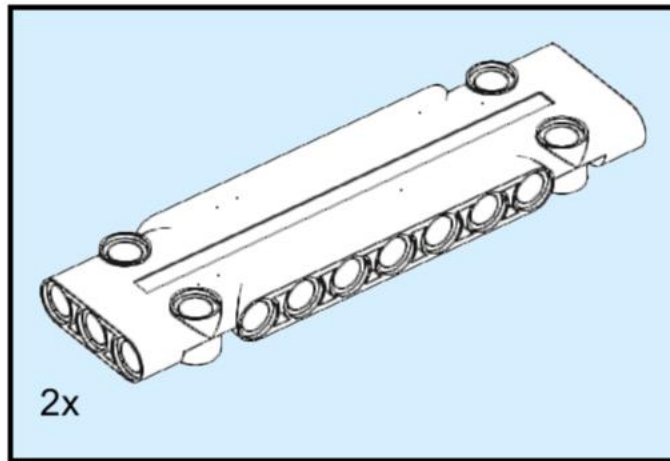
# 30



# 31

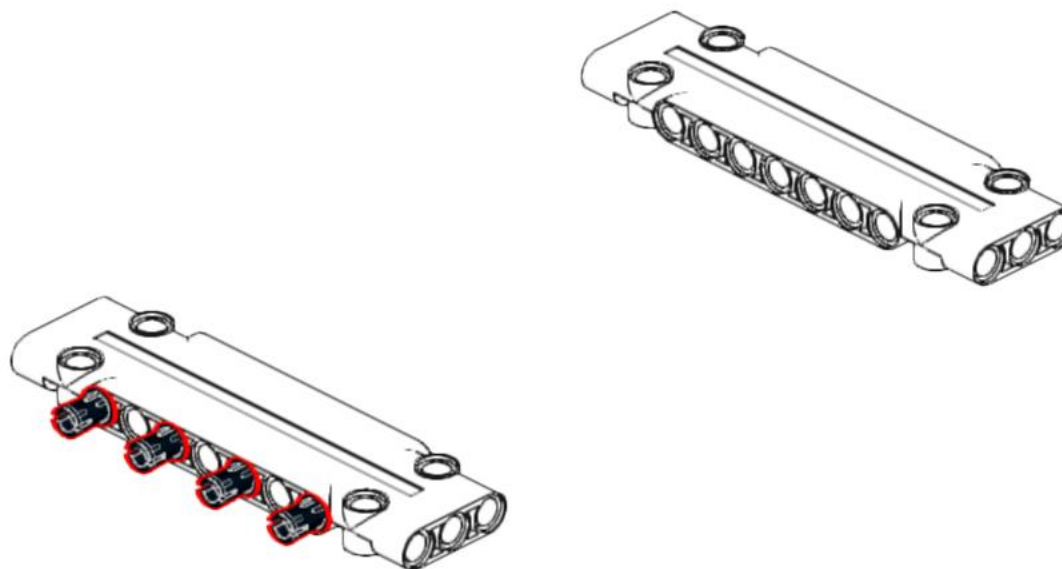


# 32

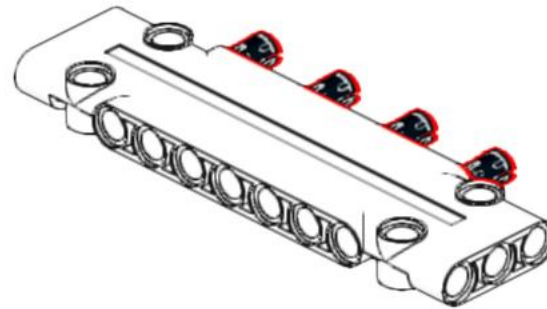
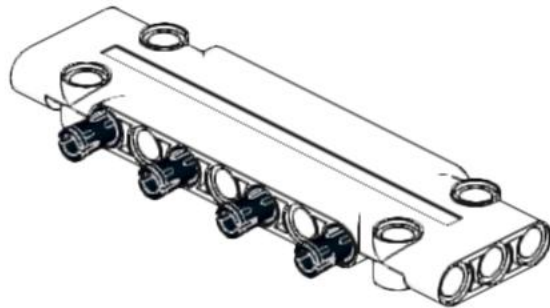




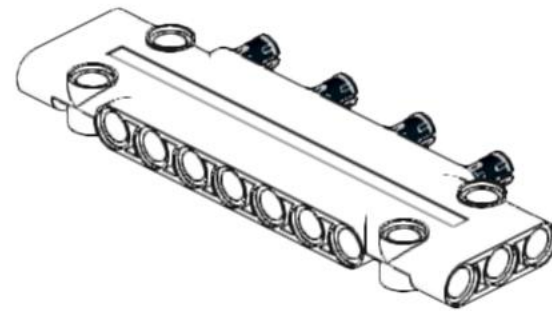
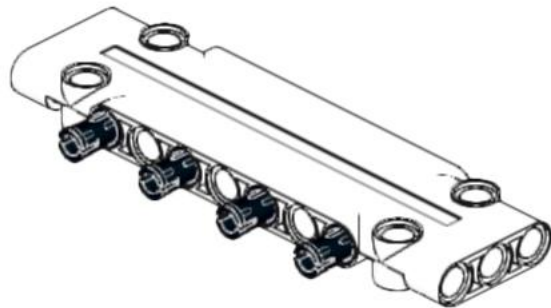
# 33



# 34

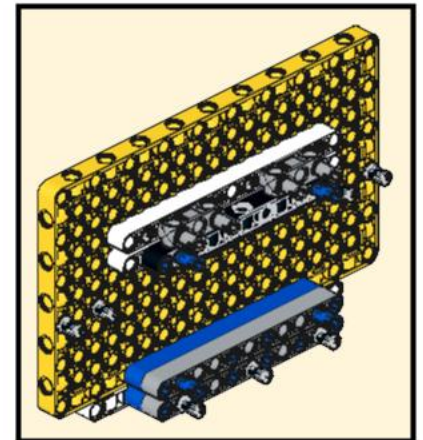
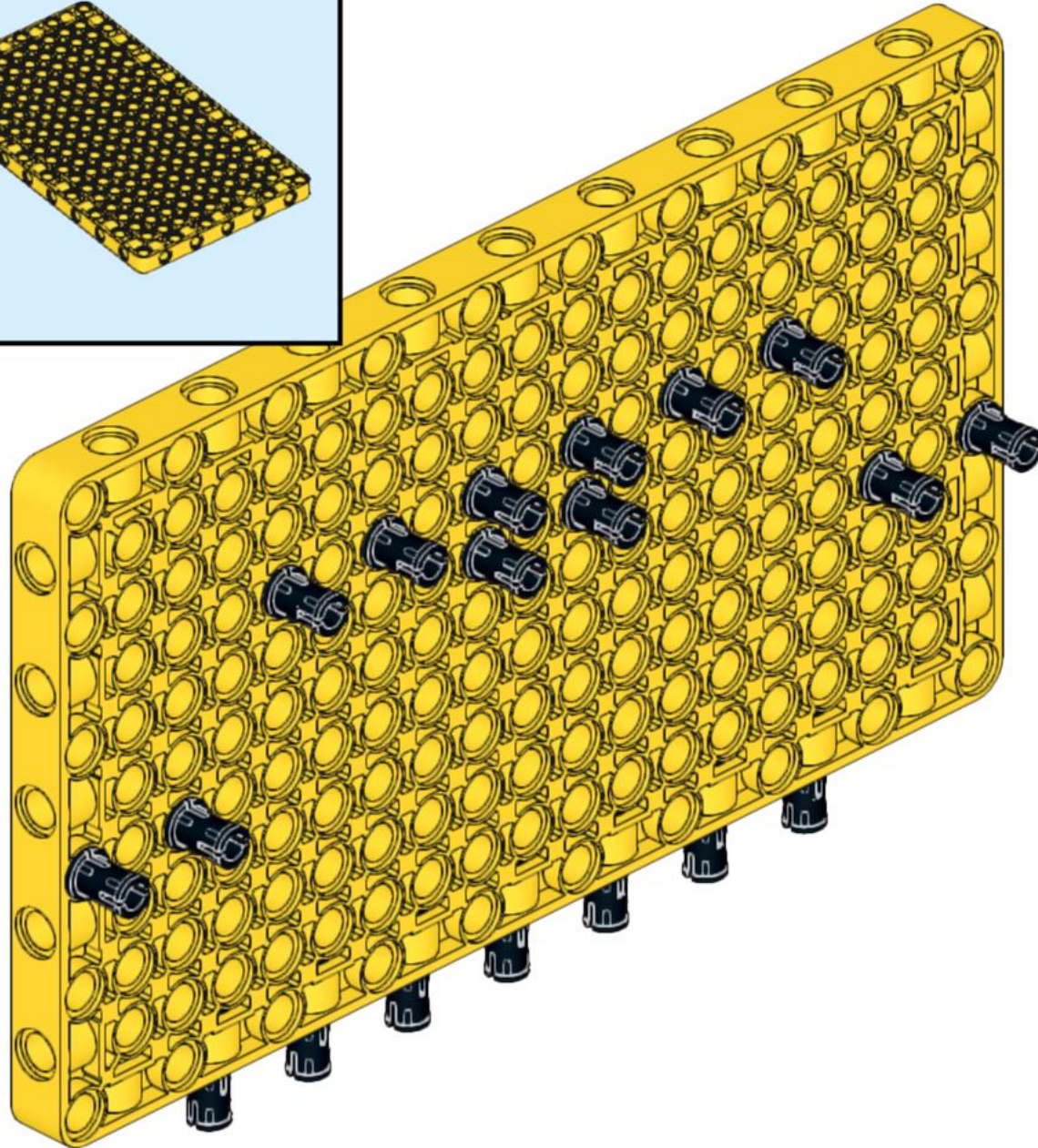
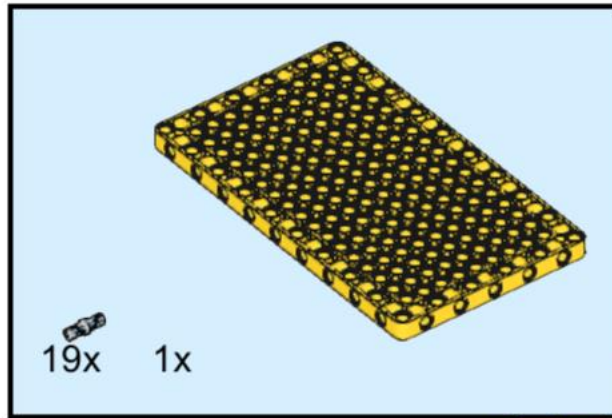


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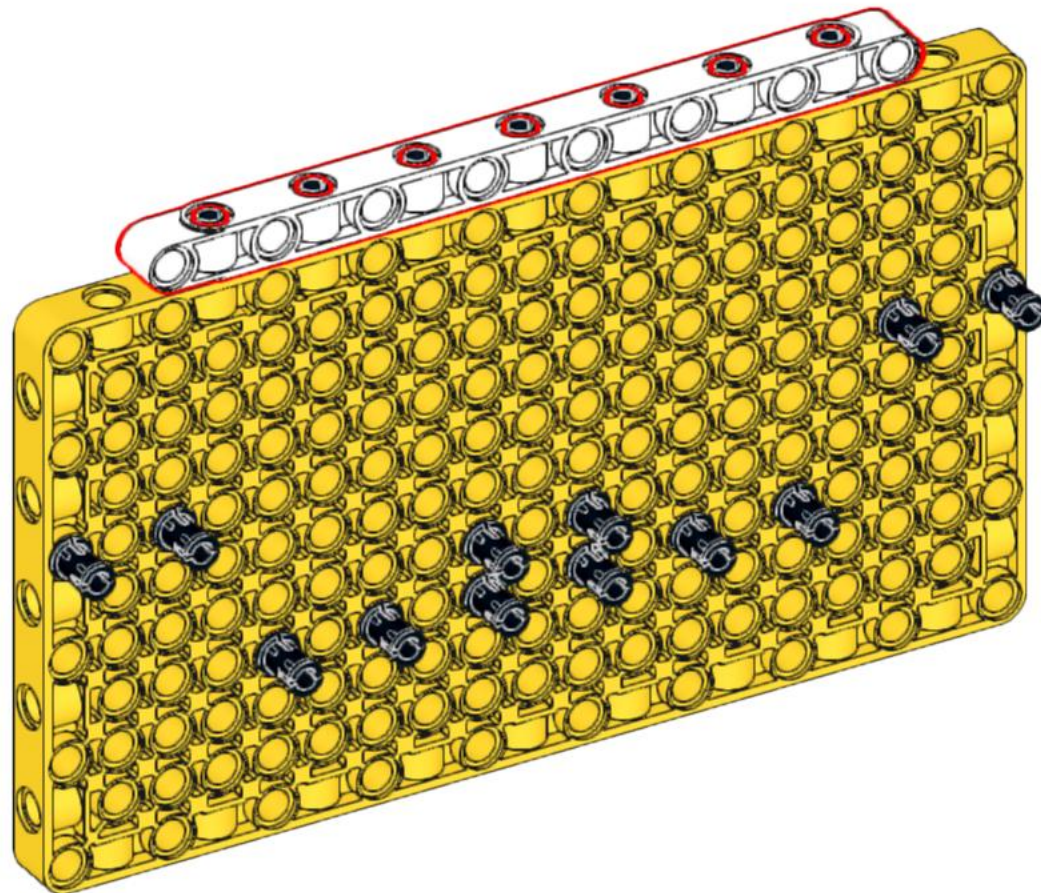
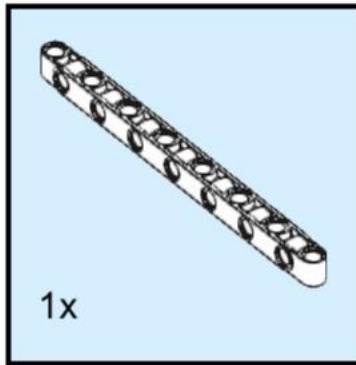




# 36

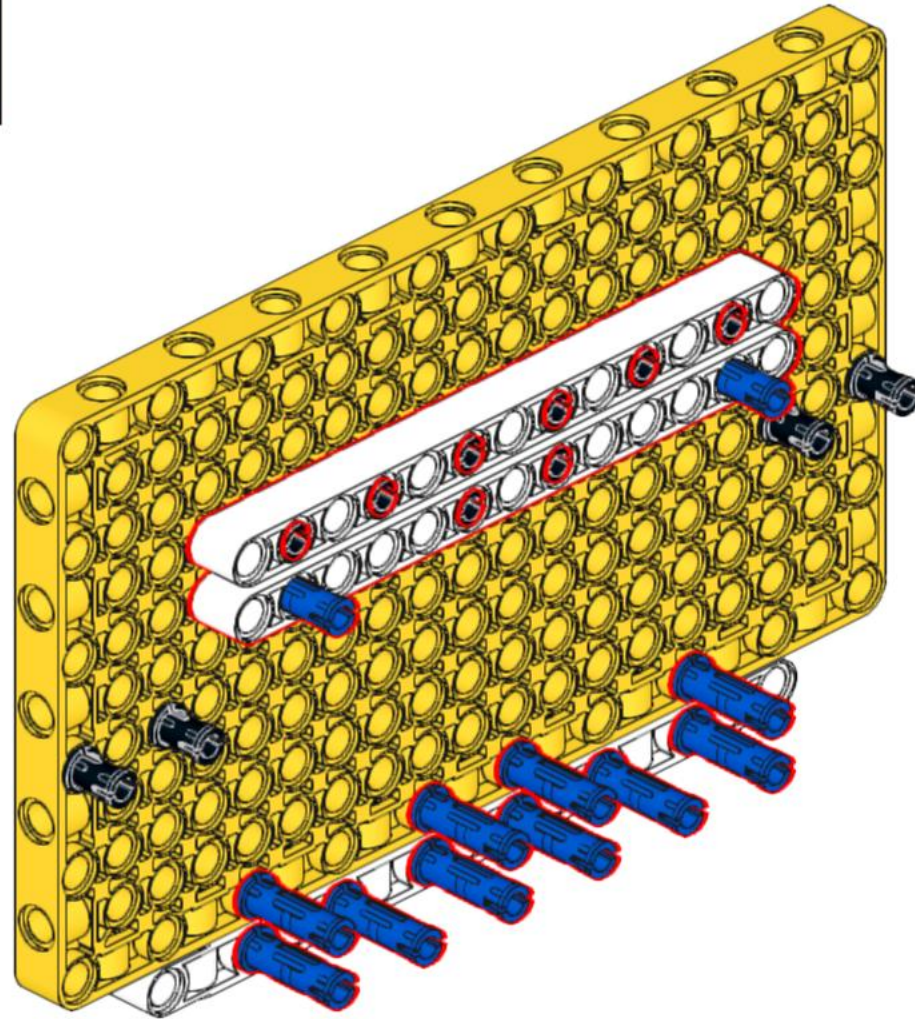
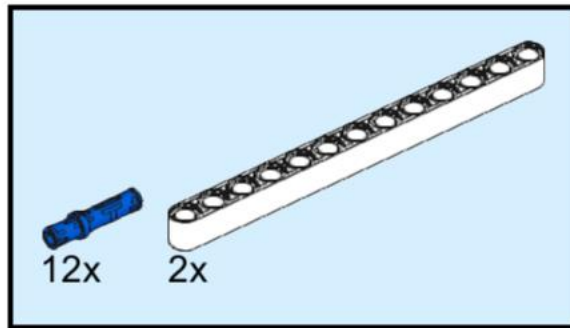


# 37



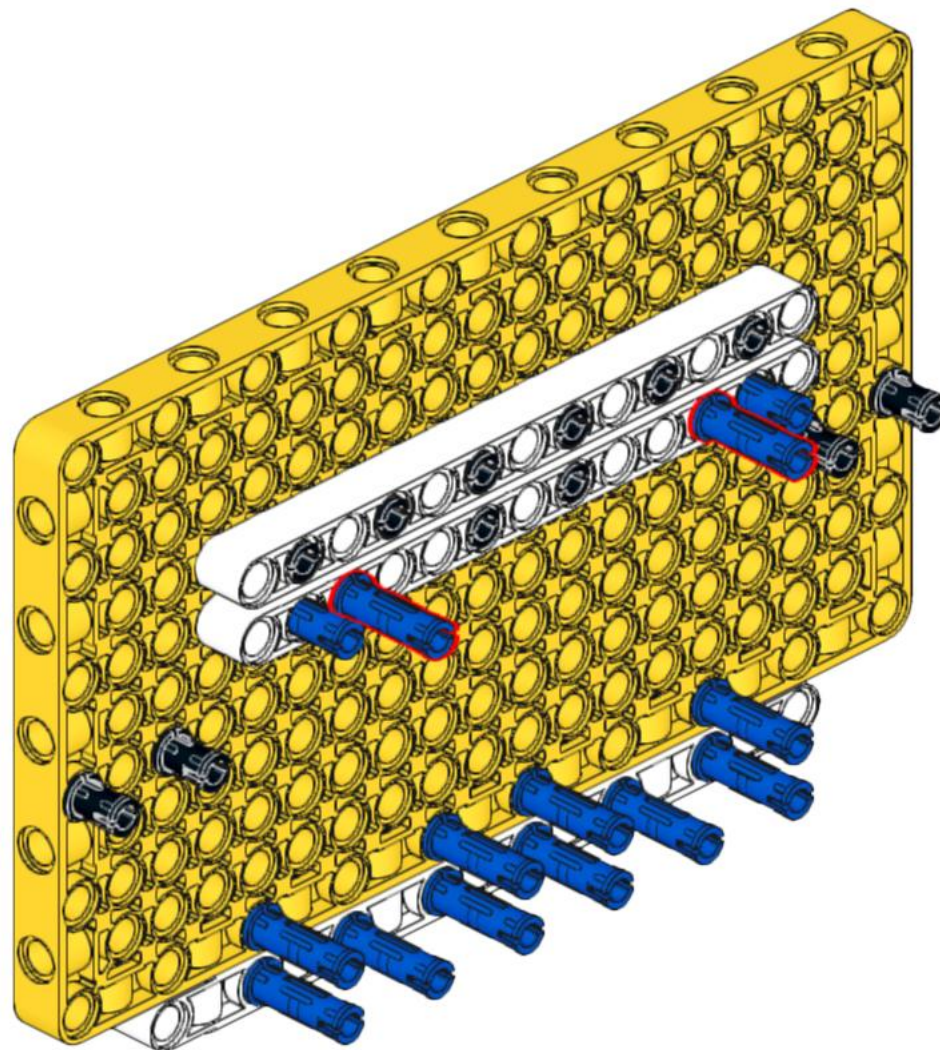
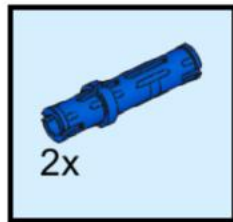


# 38

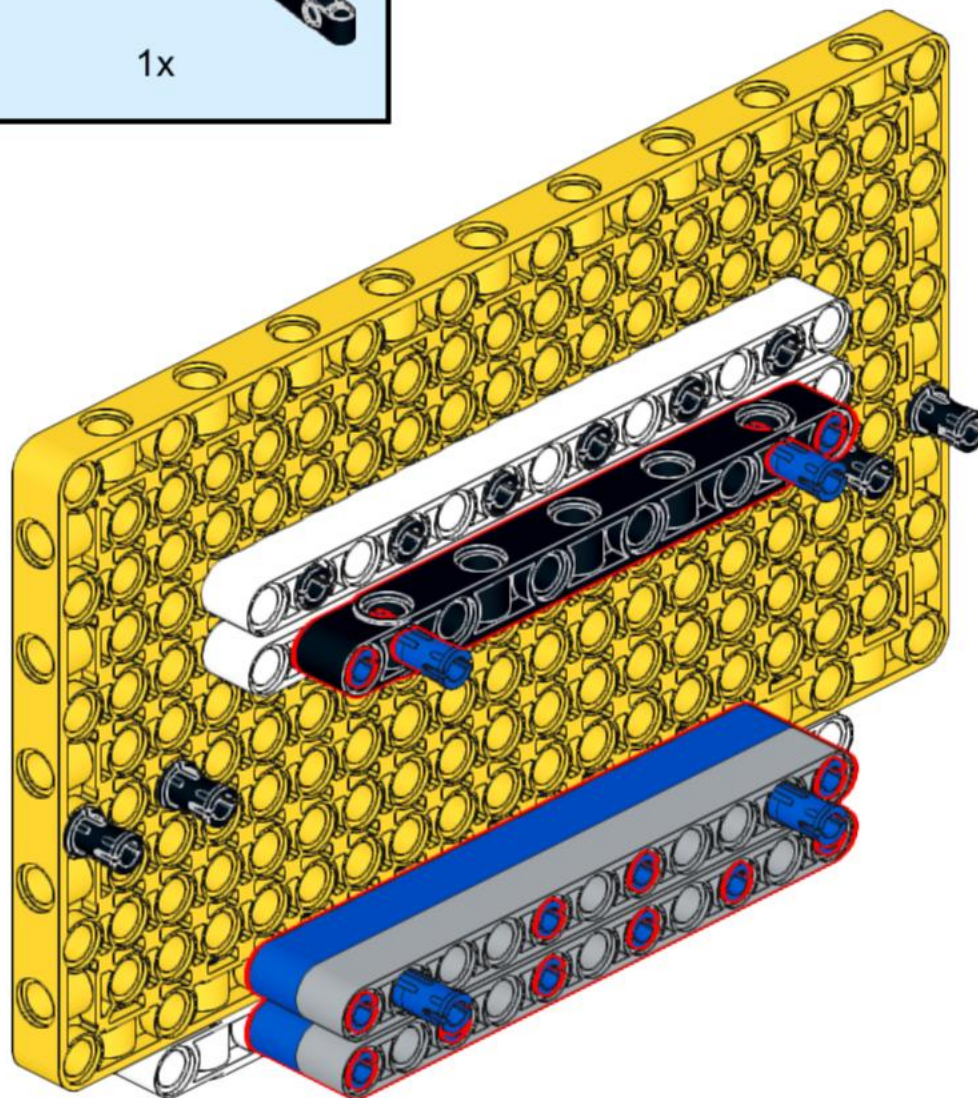
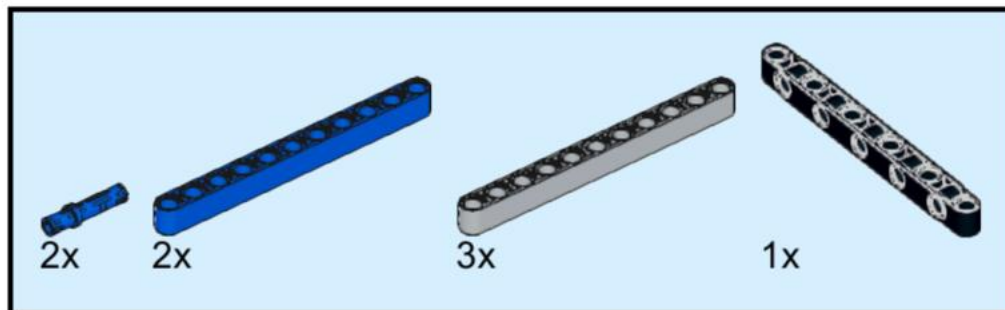




# 39

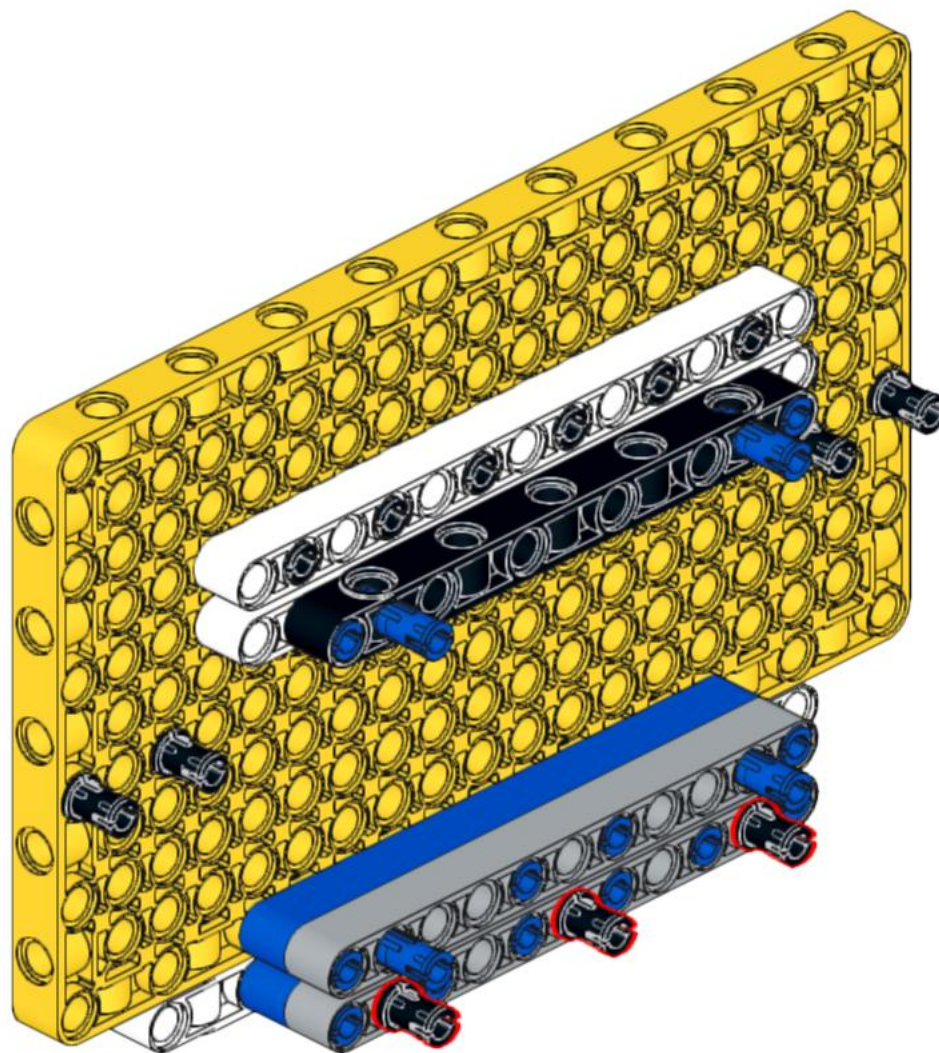


# 40



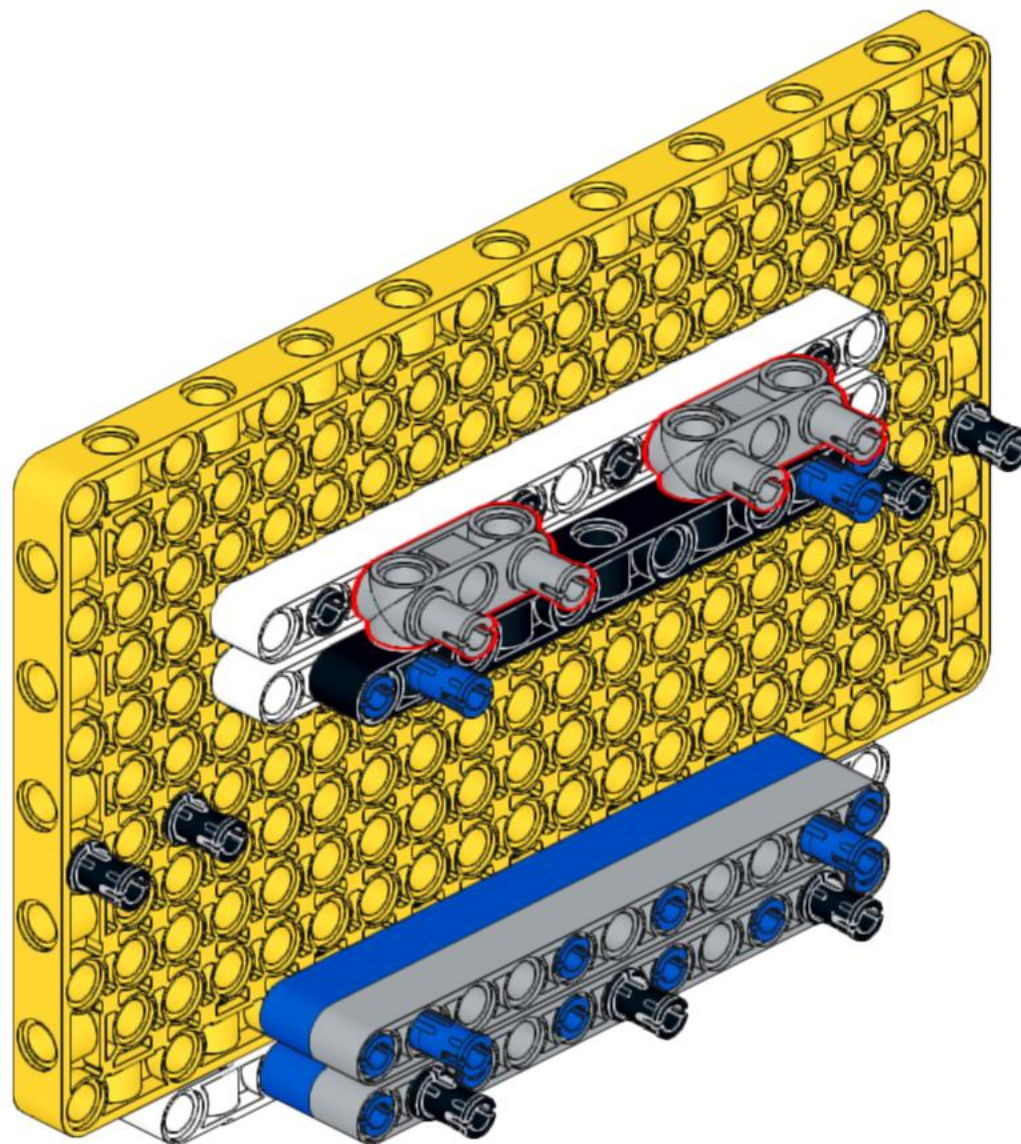
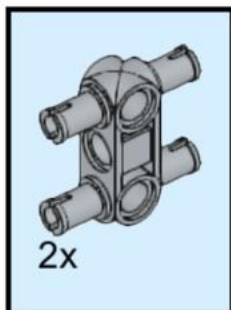


# 41

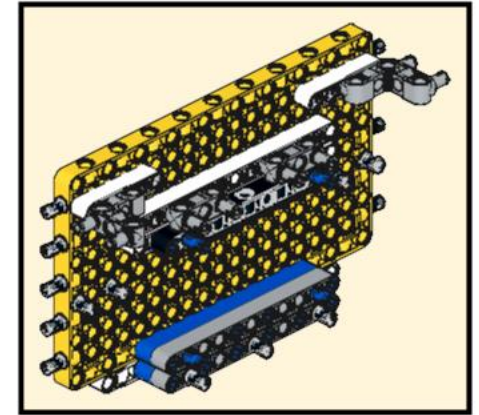
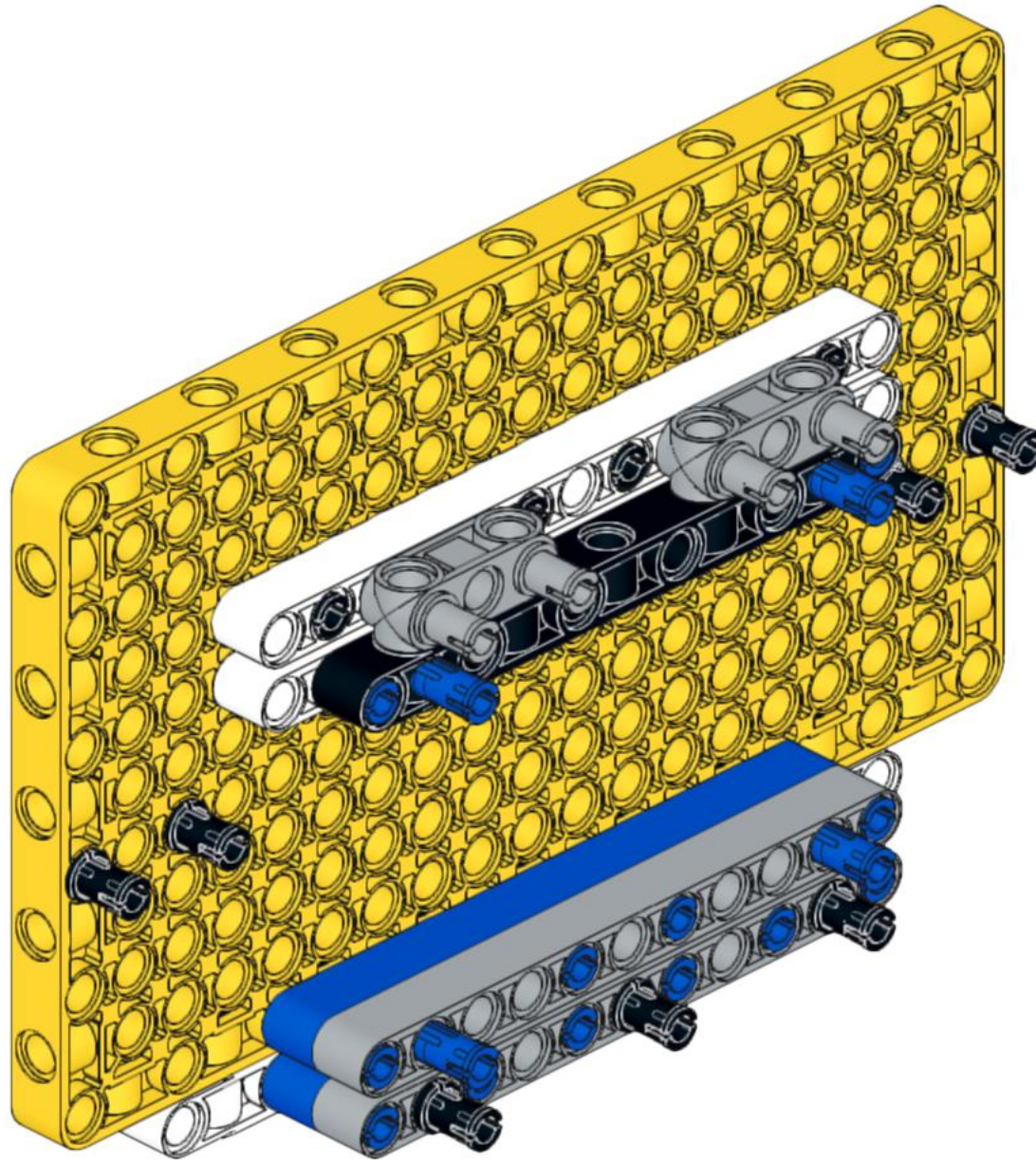




# 42

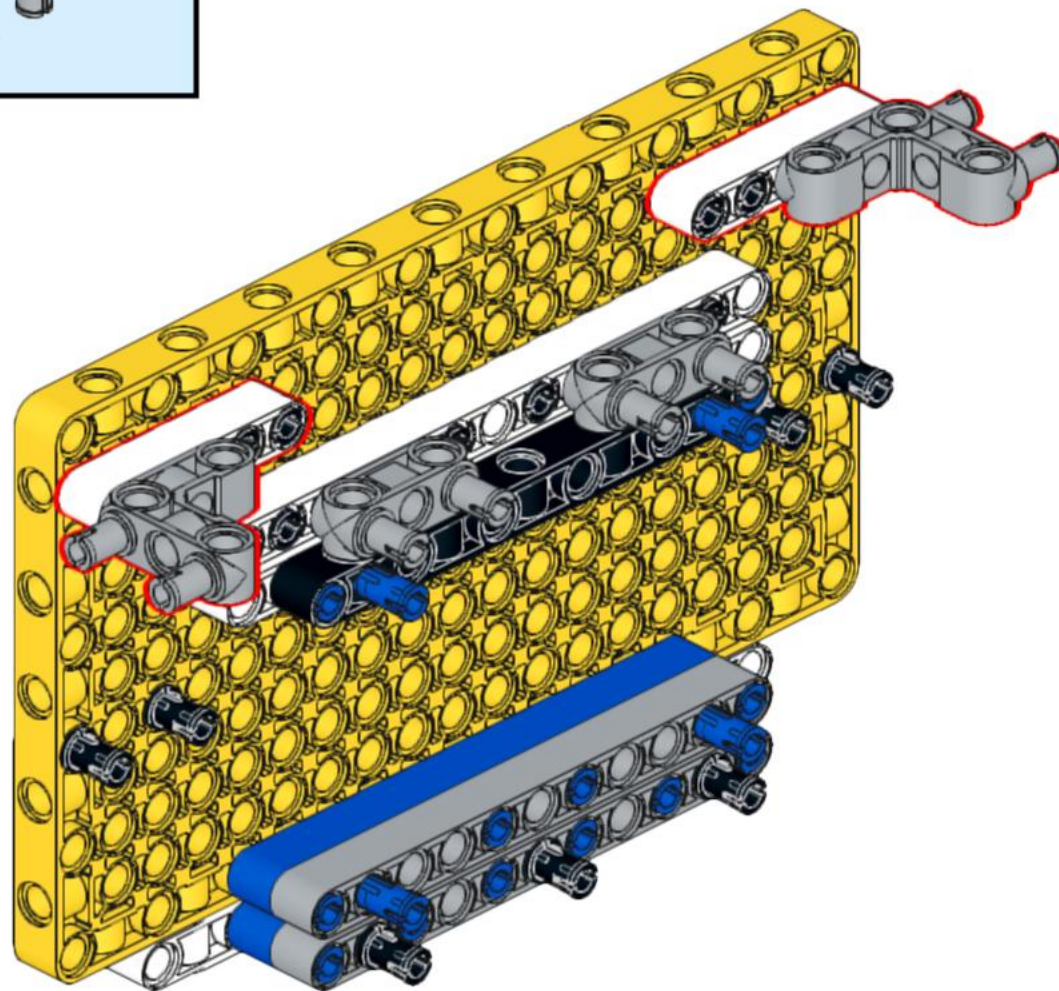
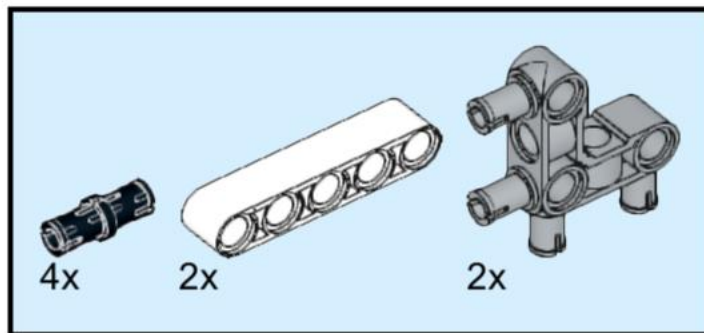


# 43



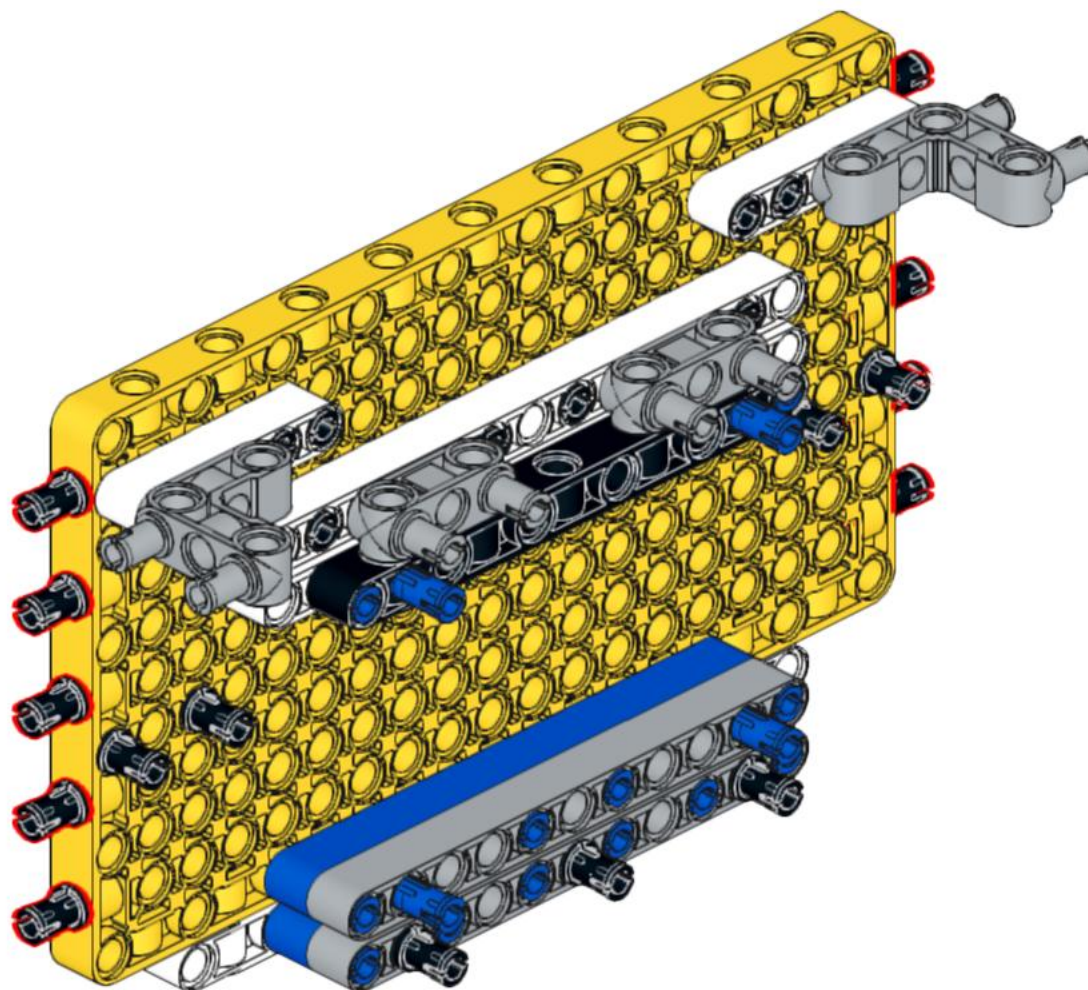


# 44

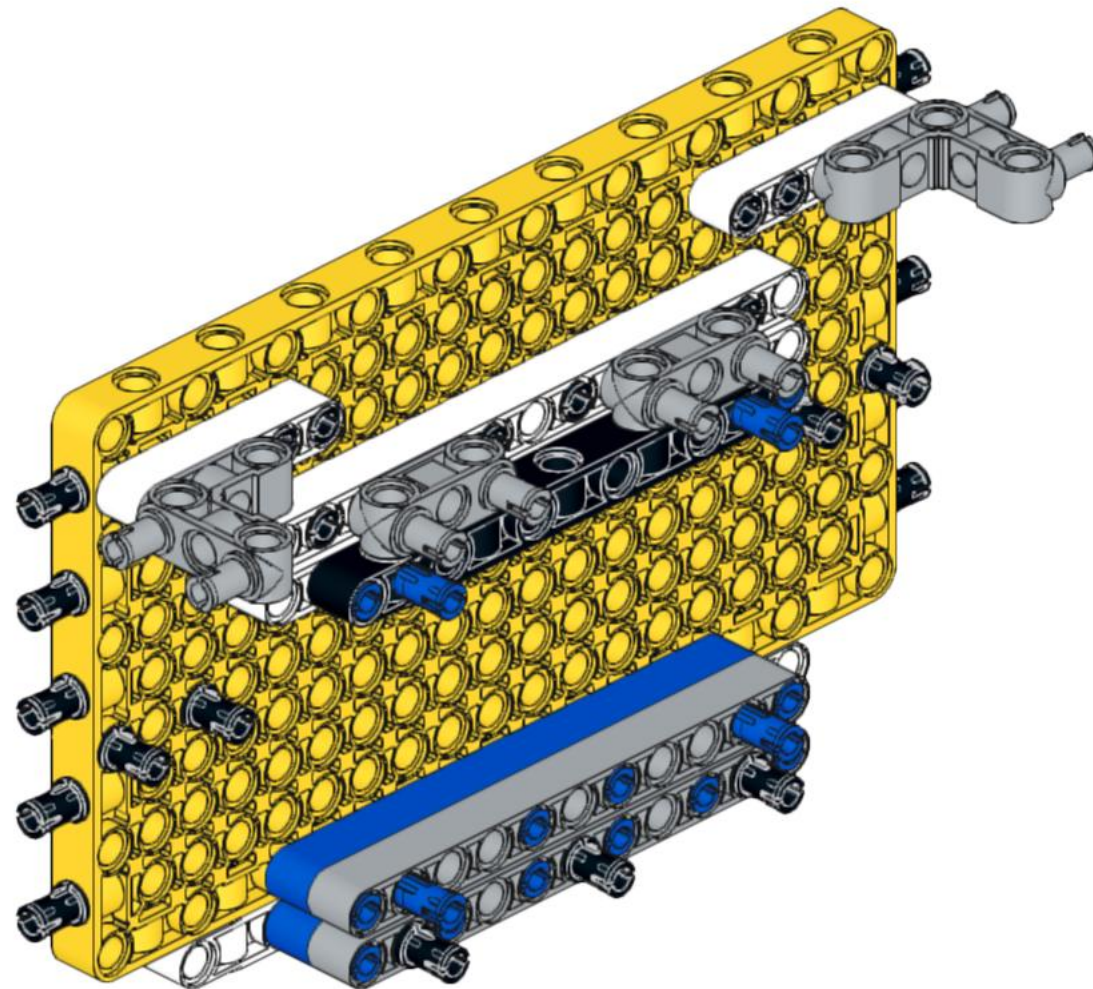




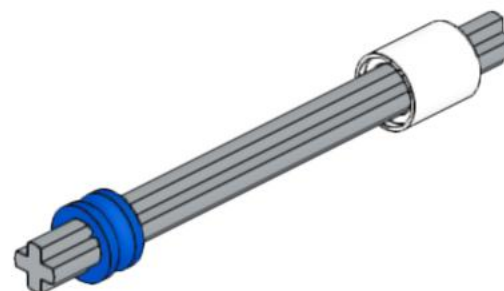
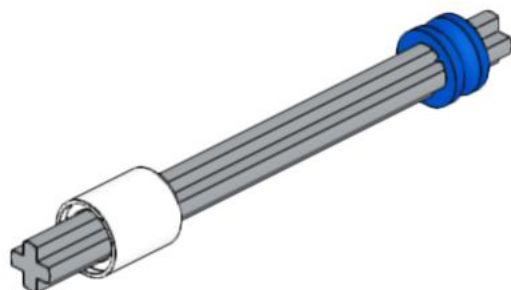
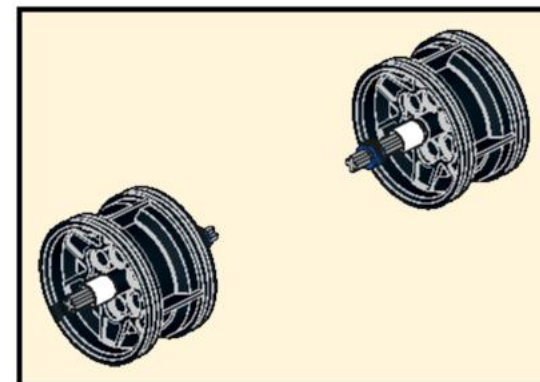
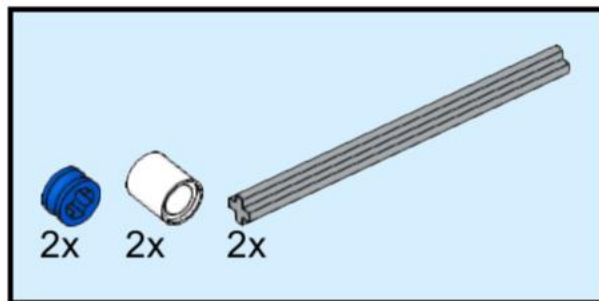
# 45



# 46

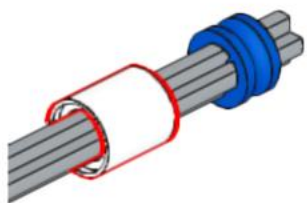
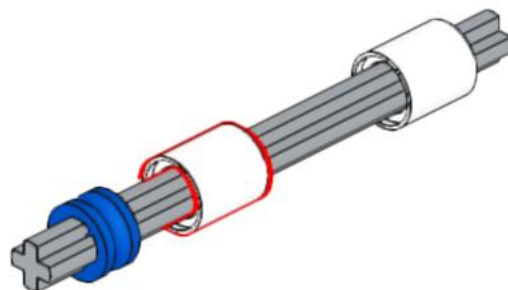
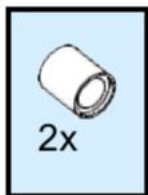


# 47

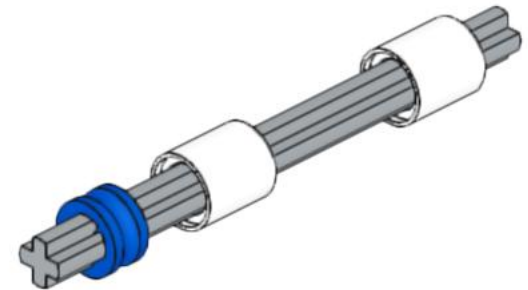
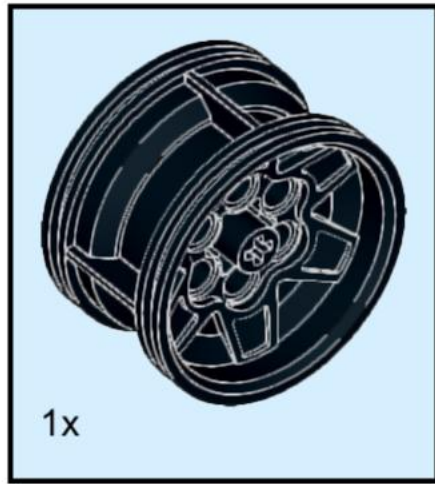




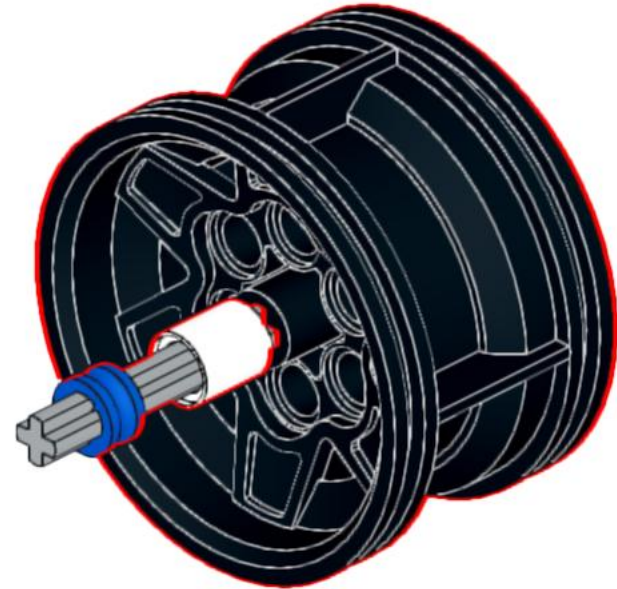
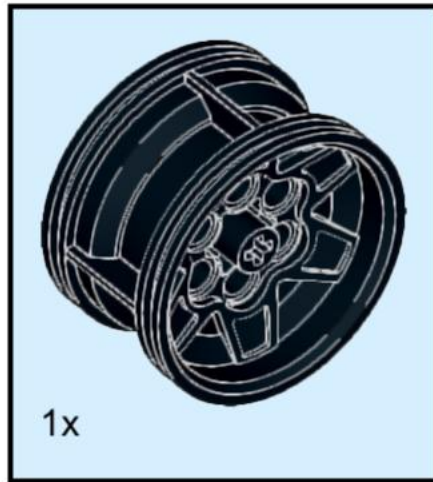
# 48



# 49



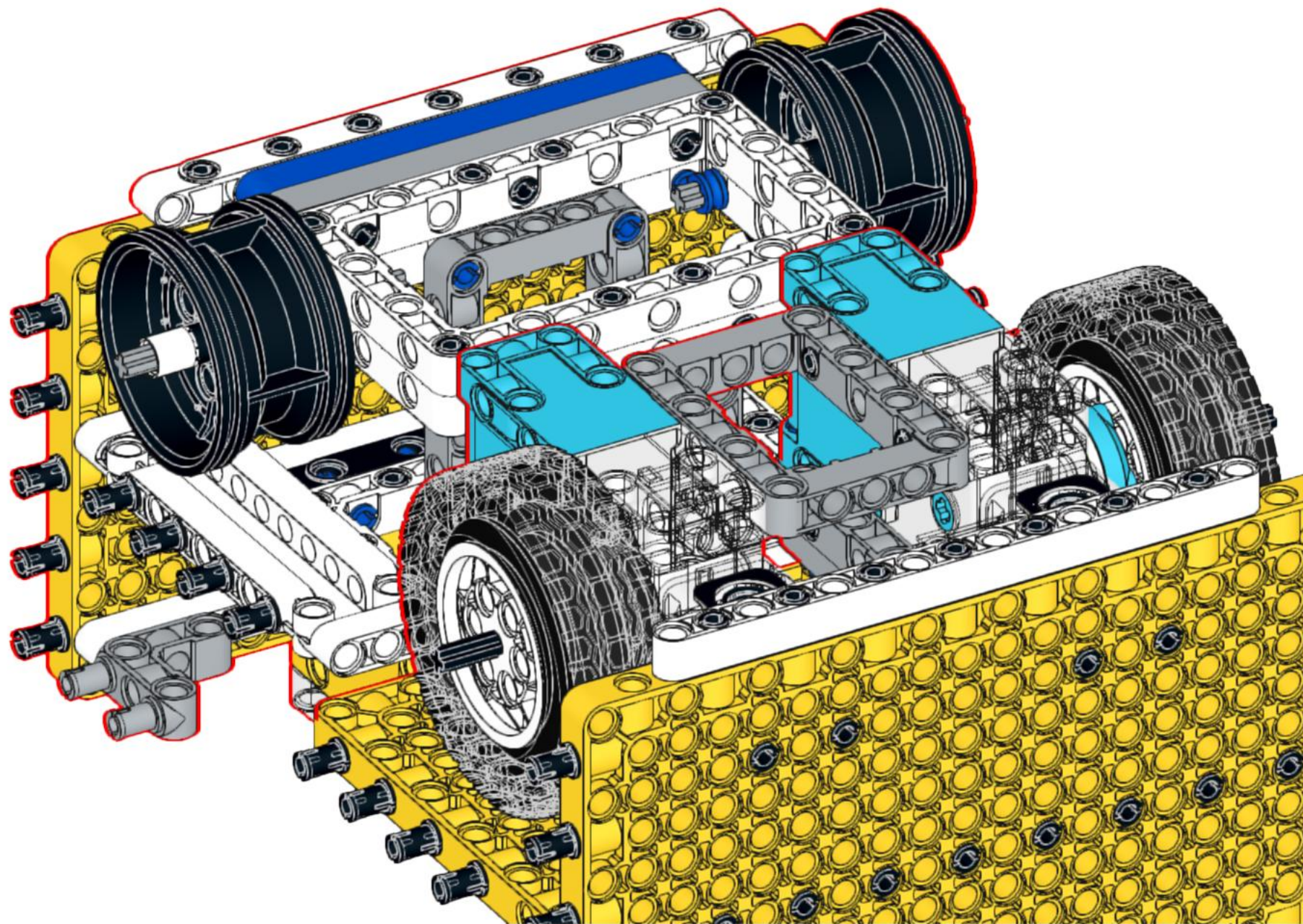
# 50



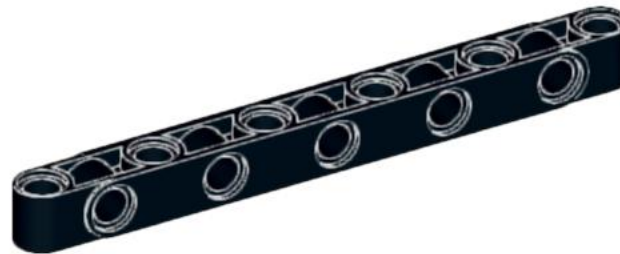
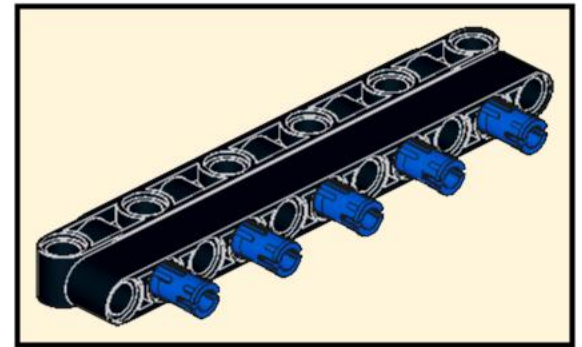
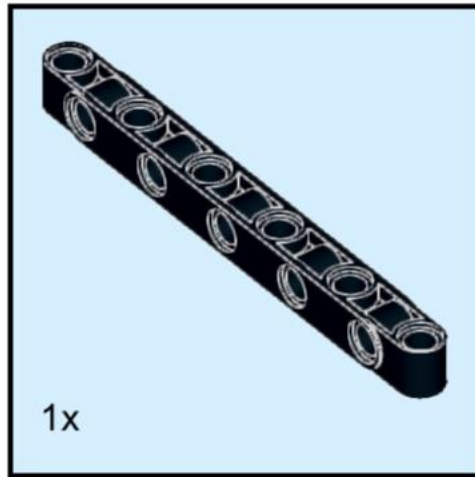
Hopefully it is obvious that these week assemblies will have be taken apart to install in the next step.



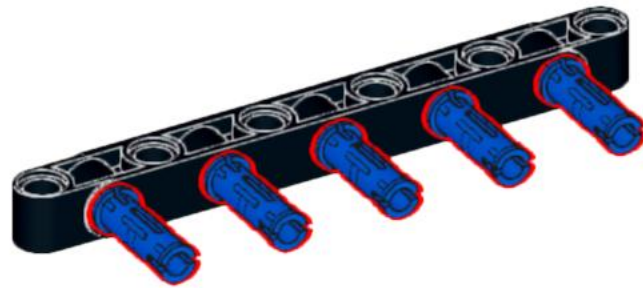
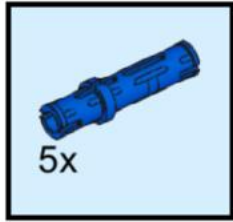
# 51



# 52

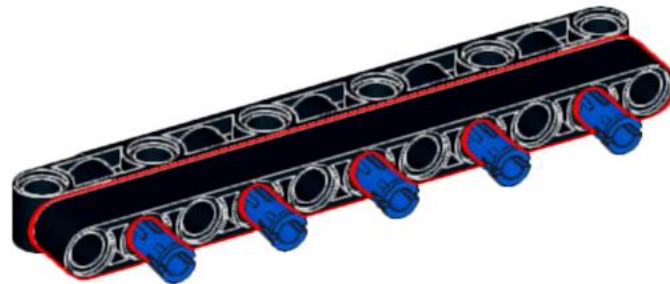


# 53

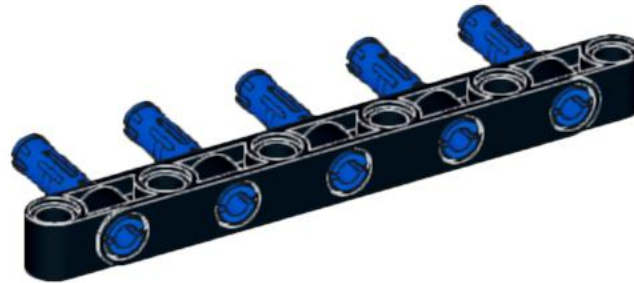
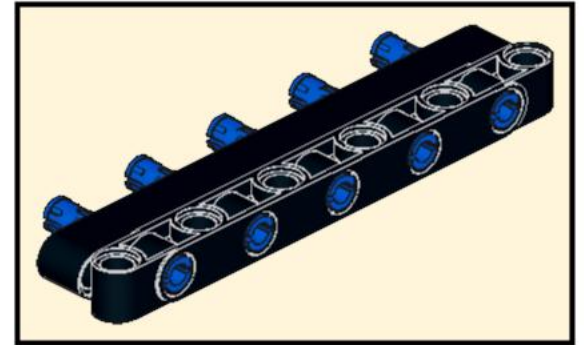
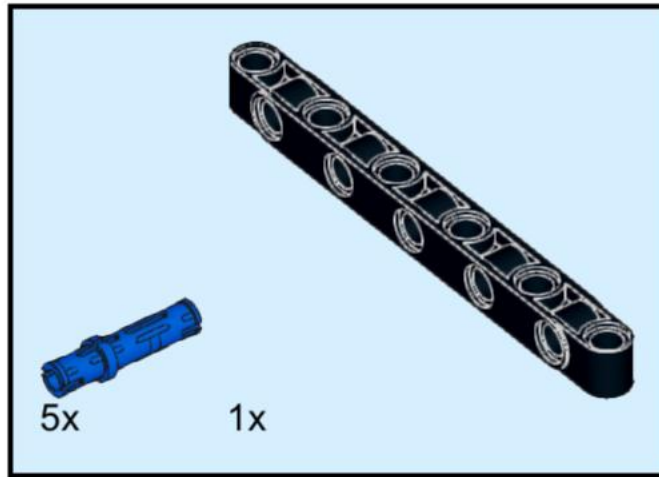




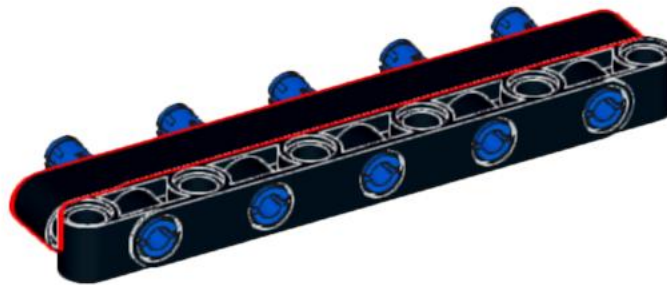
# 54



# 55

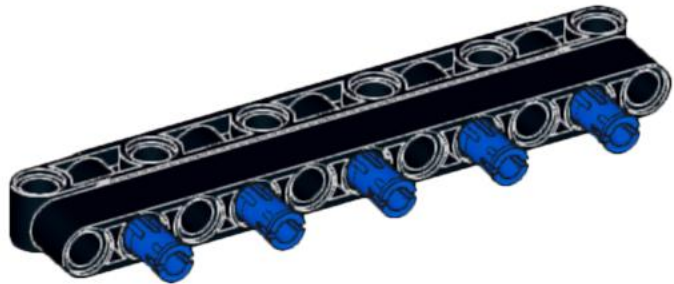
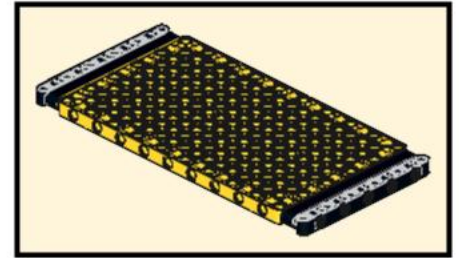


# 56

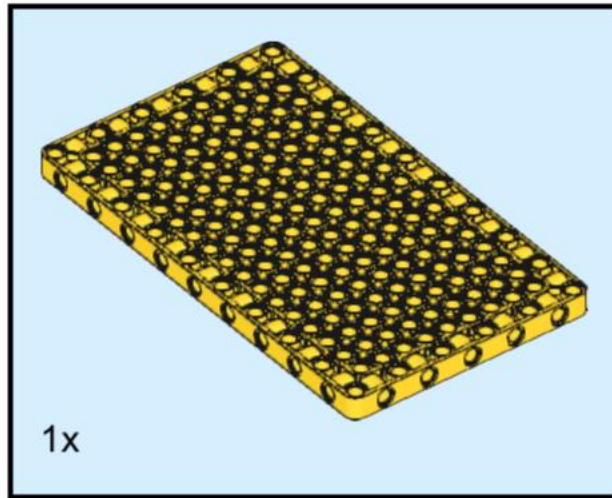




# 57

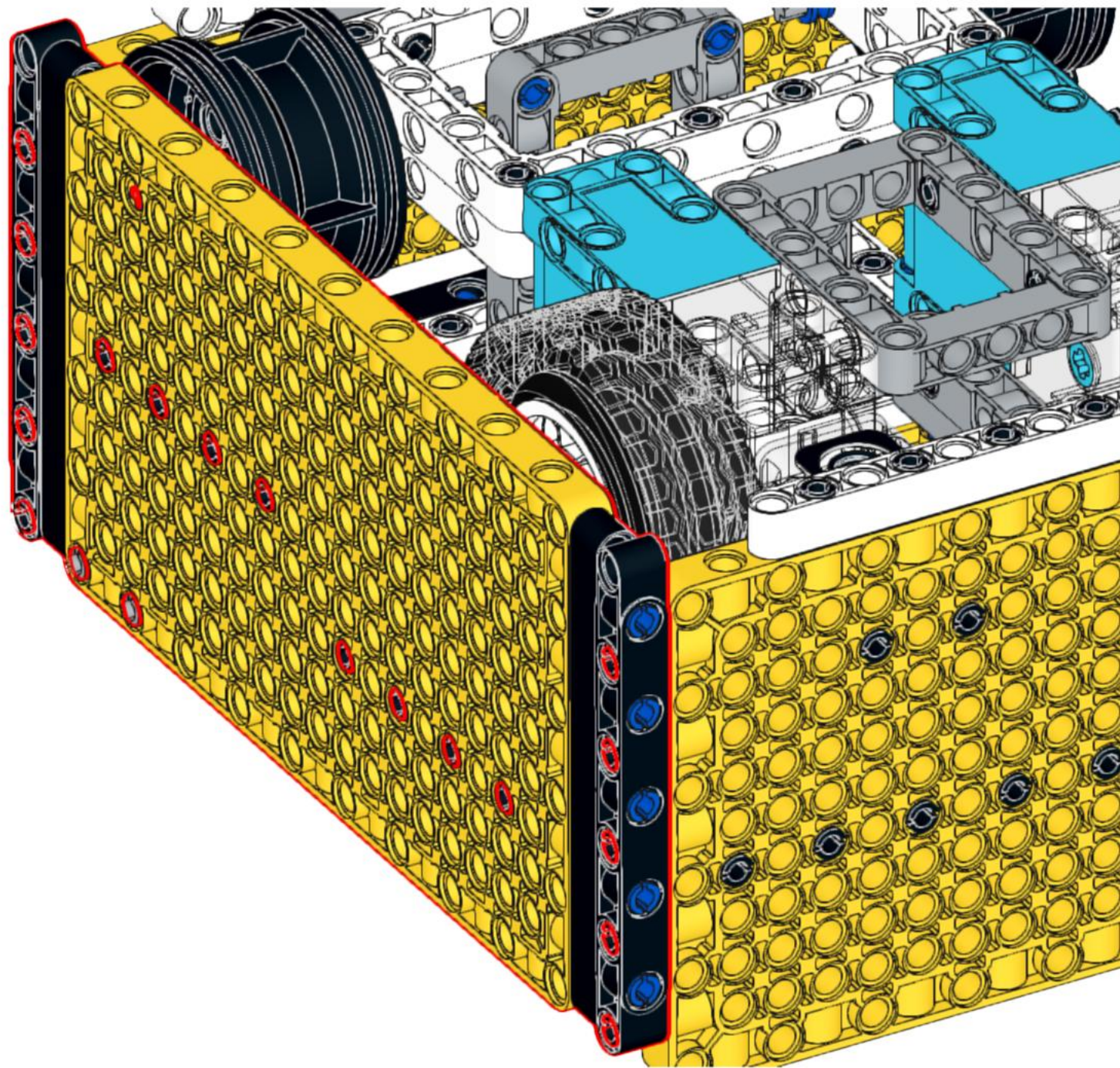


# 58



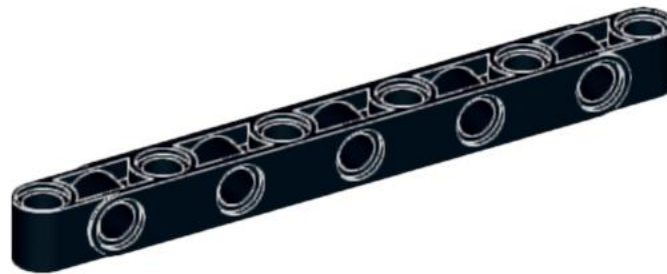
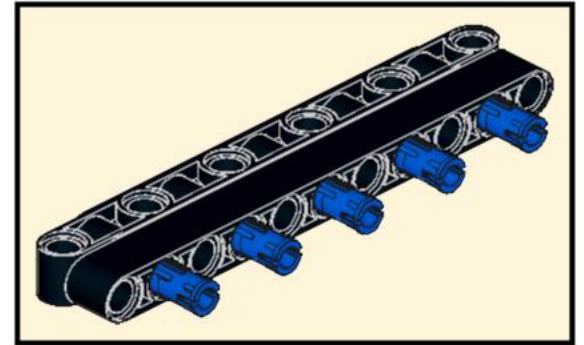
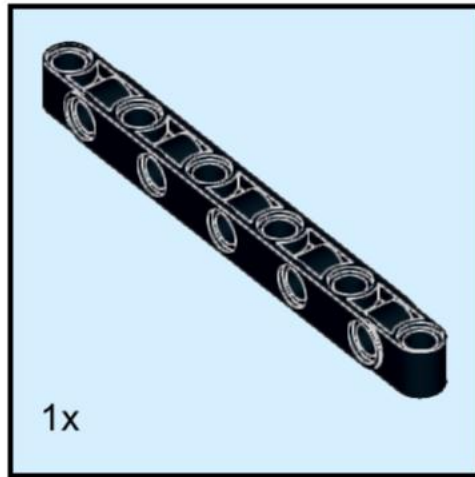


59

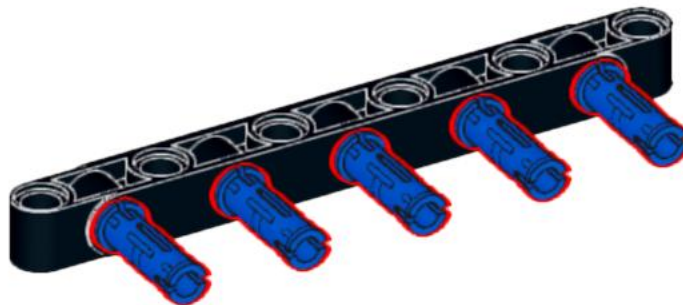
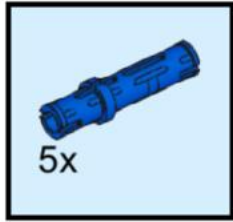




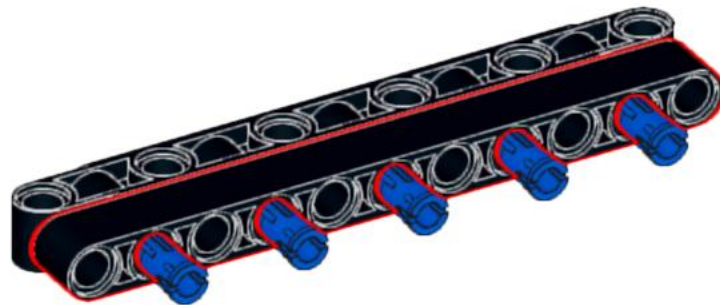
# 60



# 61

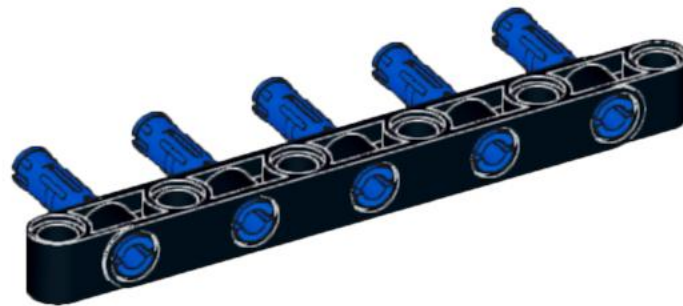
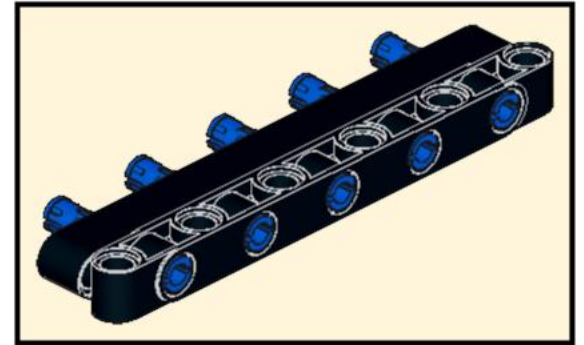
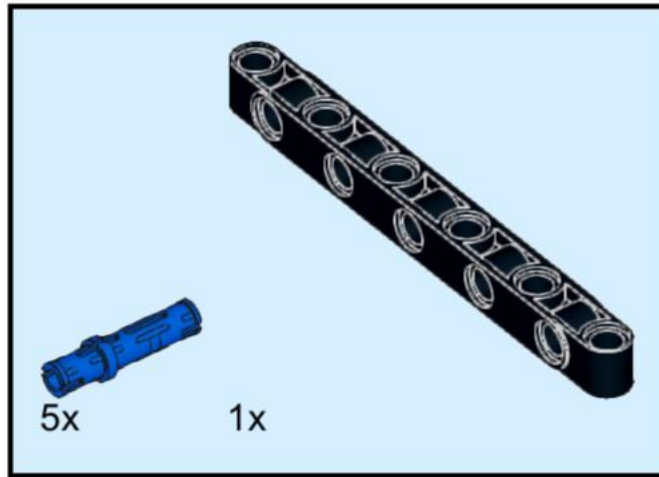


# 62

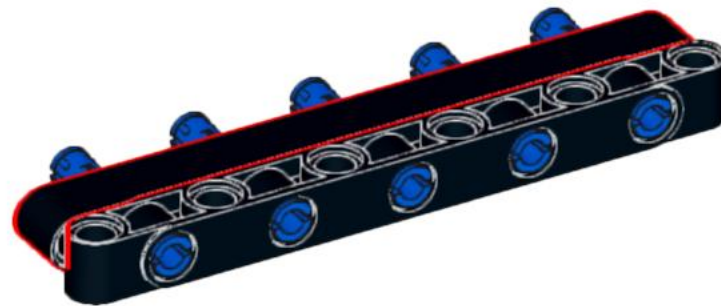




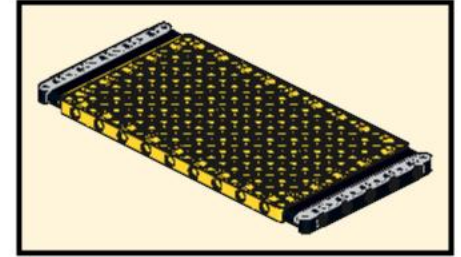
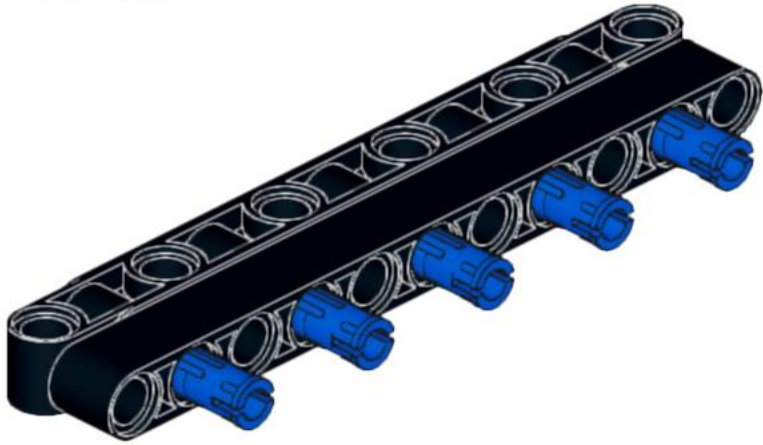
# 63



# 64

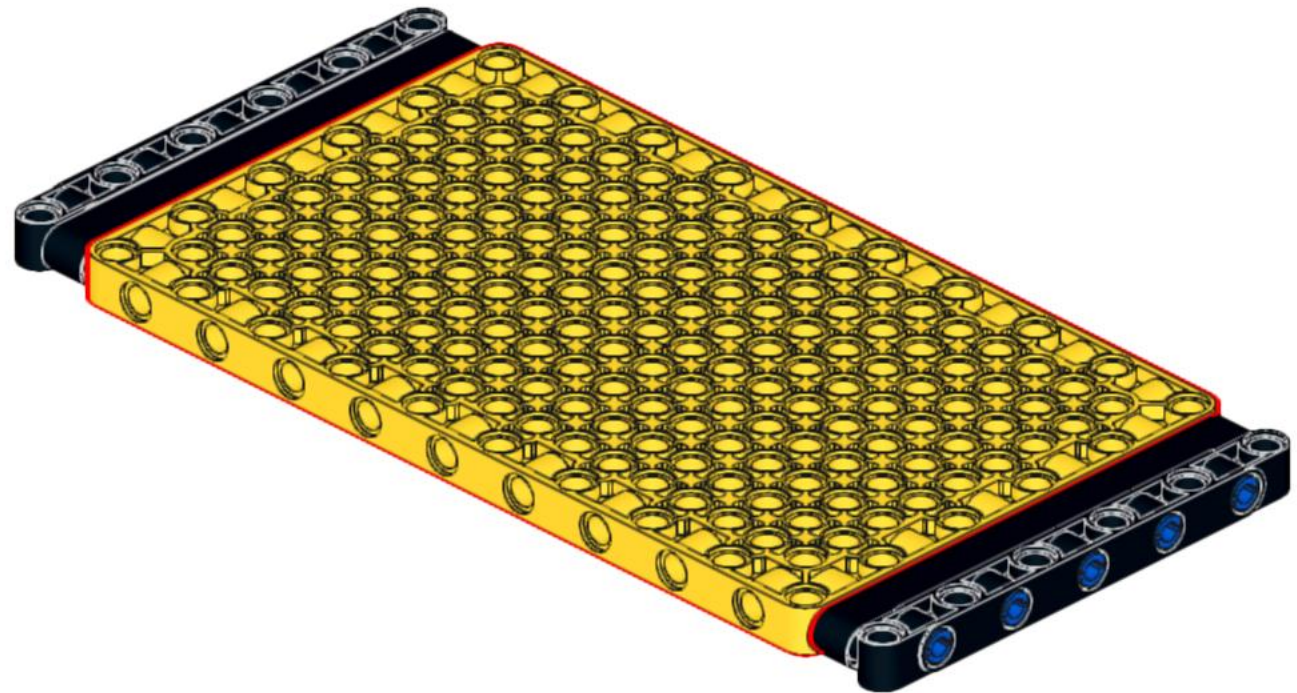
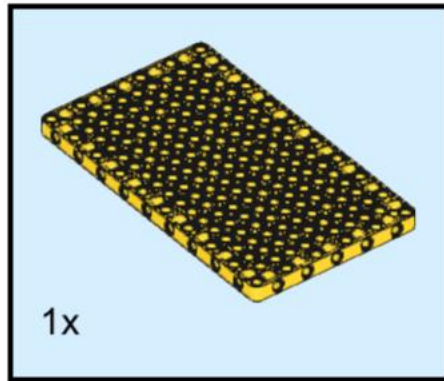


# 65



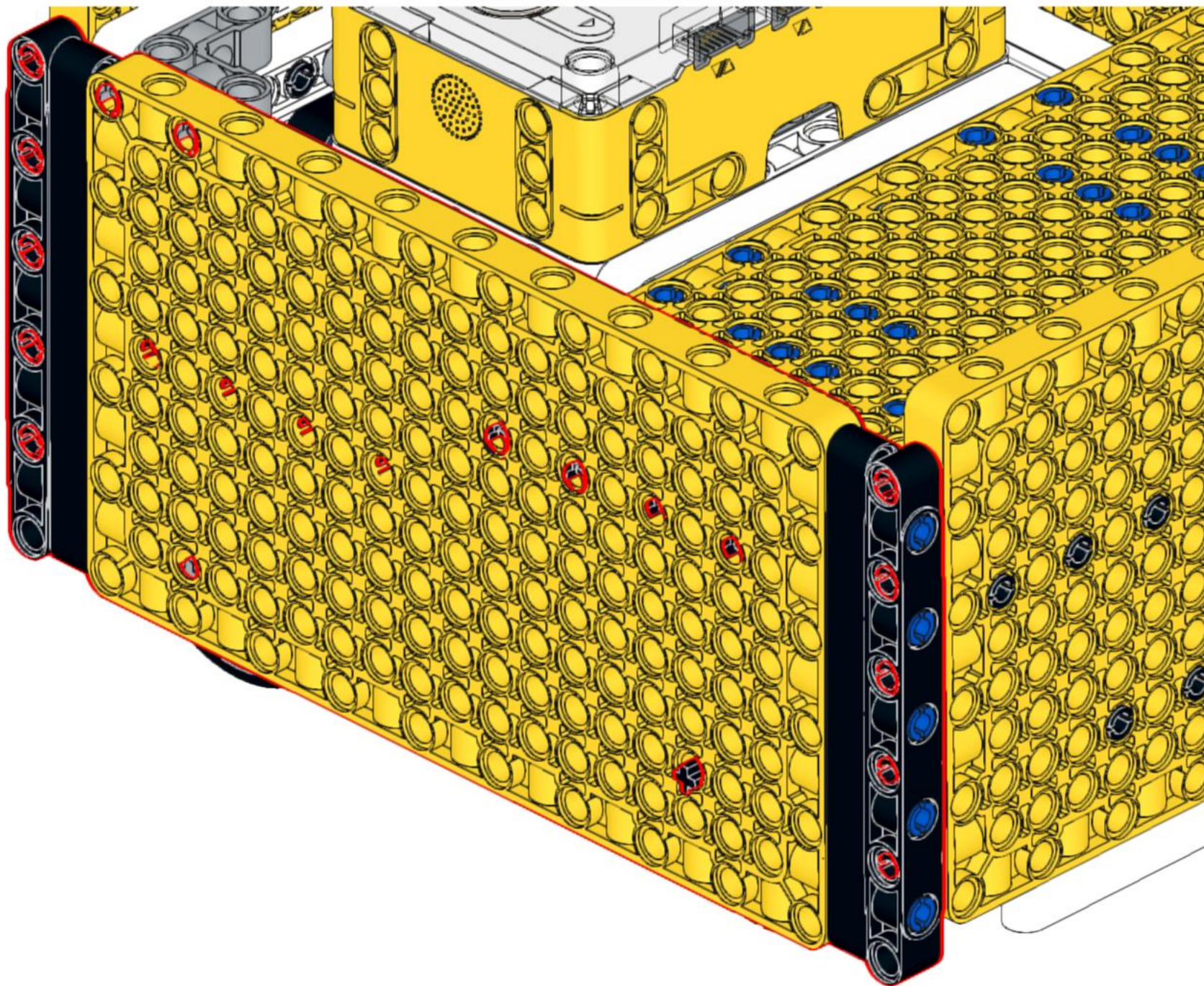


# 66



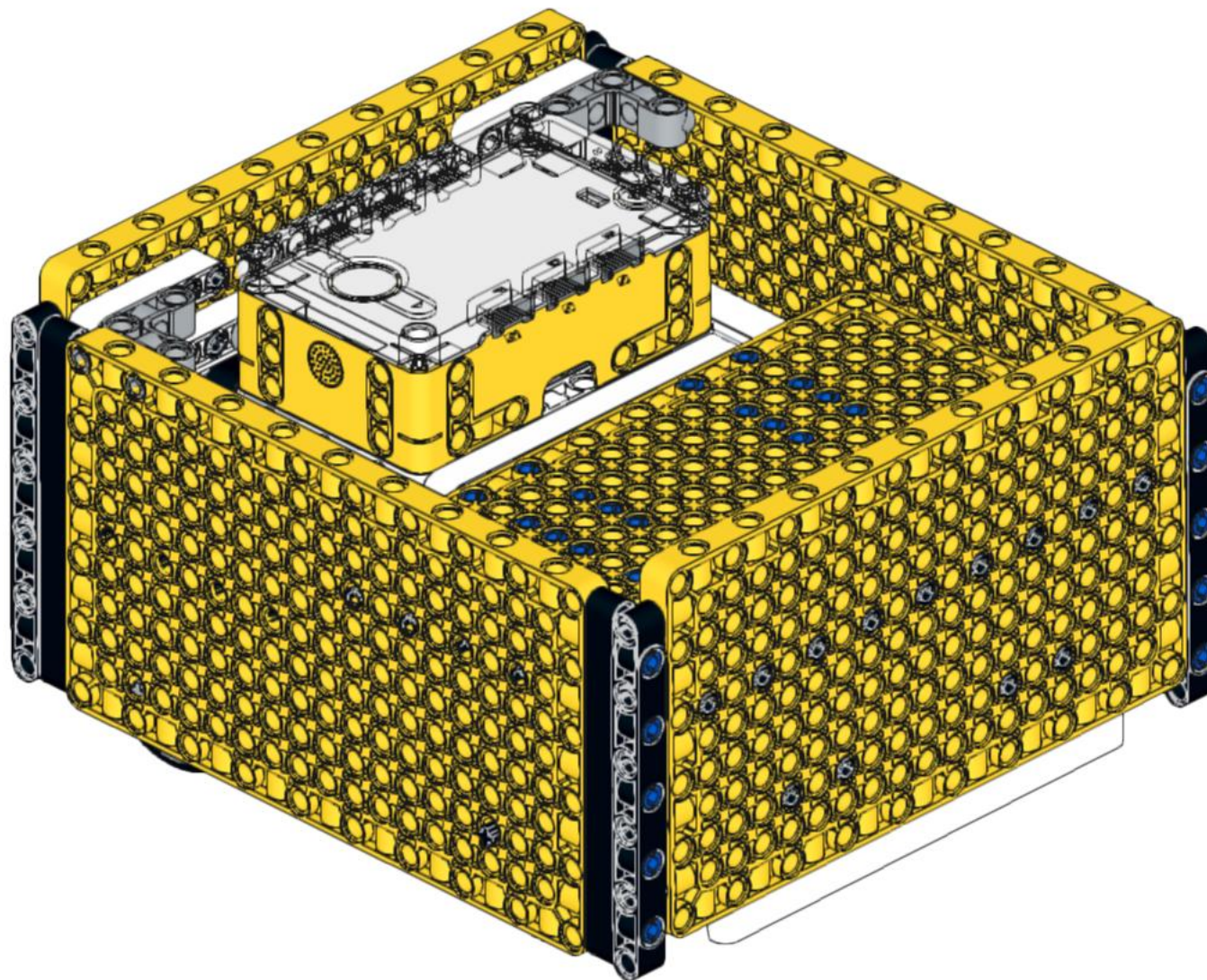


67





68





The core box robot is completed.

The next step is to install your motors. We recommend using the large Spike motors due to substantially better torque over the medium Spike motors.

There are a substantial number of ways you can install the motors based on how the team wants its attachments to work.